

Force, commitment, and perspective from Mandarin Chinese^{*}

Yuyang Liu • Yitong Luo

Yale University

Abstract

This paper focuses on the peripheral hierarchy of Mandarin interrogatives. Based on question markers' embeddability profiles in a series of nonquotational subordinating contexts, this paper argues for a three-way split in projections that contribute to the illocutionary force, attributing the non-co-occurrence of question markers to their semantics, reflecting Dayal's (2025) three-way split in the left periphery. Specifically, we propose that (i) ForceP marks the starting point of clause-typing and houses the most embeddable *wh*-operator, *meiyou*, and A-not-A; (ii) PerspP triggers perspectival centering and houses question markers restrictedly embeddable like *bu*, *ma*, and *ne*; and (iii) CommitP modifies the speaker's commitment and houses the unembeddable *ba*. Looking deeper into the embeddability patterns of PerspP-level question markers, we argue that our PerspP should be interpreted à la Dayal in the sense that it introduces a not-at-issue requirement for the embedded question to be potentially active for the matrix subject. We explicate that the embedding of PerspP-level question markers is restricted by two constraints: (i) the embedded clause size must satisfy the syntactic selectional requirement of the embedding context; and (ii) the overall interpretation must be compatible with a not-at-issue requirement lexically introduced. This proposal calls for a theory of selection that manifests a conspiracy of syntax and semantics. Moreover, we fully flesh out the semantic account for the non-co-occurrence of question markers by arguing for two fundamental distinctions in their semantic representations: one between assertions and questions and the other between monopolar and bipolar questions. We then demonstrate the need to allow for clause-typing to take place above ForceP, suggesting that semantic transformations in the left periphery are not mapped to specific syntactic projections but are only specified for their minimal structural domain.

Keywords question marker • sentence-final particle • left periphery • illocutionary force phrase • perspective phrase • monopolar question • clause-typing • Mandarin Chinese

^{*} We owe our deepest gratitude to Veneeta Dayal for her advice. We are also grateful to Shifan Chen, Yulin Du, Lingxiao Gao, Xi Huang, Zhixian Huang, Manxin Lan, Yiming Li, Yunhao Li, Chengwei Liang, Nianpo Su, Ziqin Wei, Zhenrui Zhang, Yi Zhou, Zhenghao Zhou, Jiamu Zhu, Xiaomeng Zhu, and twelve anonymous survey participants for Mandarin data and judgments. For invaluable discussion and feedback, we thank Athulya Aravind, Carla Bombi, Mirium Butt, Simon Charlow, Qiushi Chen, Kathryn Davidson, Daniel Hole, Miao-Ling Hsieh, Xuhui Hu, Magdalena Kaufmann, Young-Hoon Kim, Richard Luo, Paul Portner, Maribel Romero, Zoltán Szabó, Milena Štreikaitė, Sze-Wing Tang, Alexander Wimmer, Jim Wood, Danfeng Wu, Yimei Xiang, Yūsuke Yagi, Ka-Fai Yip, Xuetong Yuan, Florence Zhang-Yukun, four anonymous GLOW 47 reviewers, and our audience at UConn Linguistics Graduate Roundtable 14, Yale Semantics Reading Group, TEAL 14, and GLOW 47. All errors are our own.

Contents

1	Introduction	3
2	Traditional view	5
2.1	Cartography of the left periphery	7
2.2	Dichotomy in question marker embeddability	8
3	New perspective	10
3.1	Trichotomy in question marker embeddability	11
3.2	Question markers with <i>meiyou</i> -type embeddability	14
3.3	Question markers with <i>ma</i> -type embeddability	15
3.4	Questions that do not embed	17
4	Conspiracy of syntax and semantics	18
4.1	Syntactic component: Split iForceP	18
4.2	Semantic component: P-activeness	19
4.3	Necessity for both components	21
4.4	Derivations of the question markers' (un)embeddability	22
5	Different wines in different vessels	24
5.1	Assertiveness and nonassertiveness	24
5.2	Applying assertiveness diagnostics in Mandarin	27
5.3	Singleton and doubleton	30
5.4	B-contexts and more in Mandarin	32
6	Clause-typing above ForceP	35
6.1	Hindi/Urdu polar question particle	35
6.2	Minimal structural domain	37
7	Conclusion	38

1 Introduction

Interrogative clauses vary in their ability to be embedded. In English, for instance, while *wh*-questions without subject-auxiliary inversion and polar questions introduced by *if* or *whether* are generally embeddable as in (1), questions in the form of a declarative with rising intonation (i.e., *rising declaratives*) are generally unembeddable as in (2). Below, we use “↑↑” to represent the rising intonation typically associated with English rising declaratives.

- (1) a. I wonder/know/have forgotten what we should do.
 b. I wonder/know/have forgotten whether it was raining.
- (2) *I wonder/know/have forgotten [it was raining]_{↑↑}.

Compelling empirical observations and theoretical insights emerge when the variation in the embeddability of interrogative clauses is revealed to be more than a mere yes-or-no dichotomy. Recent work documents and theorizes a third possibility of question embeddability (Woods 2016, Dayal 2025) beyond the standard embeddable (nonroot) vs. unembeddable (root) distinction (a.o.: Emonds 1970, 1988, 2004, Hooper & Thompson 1973). In some English dialects, questions involving subject-auxiliary inversion are embeddable but constrained by independent conditions (Burling 1973, Henry 1995, McCloskey 2006) as in (3). These embedded constructions are known as *embedded inverted questions* (EIQs).

- (3) a. I wonder what should we do. (McCloskey 2006: 89)
 b. #I know what should we do.
 c. I have forgotten what should we do.

Drawing on crosslinguistic evidence, Dayal (2025) accounts for the presence of three possibilities (including EIQs) by positing a “CP < Persp(ective)P < S(peech)A(ct)P” hierarchy in the left periphery, a three-way distinction reminiscent of Coniglio & Zegrean’s (2012) “C(lause)T(ype)P < Ill(ocutionary)Force)P” and Woods’s (2016) “ForceP < I(llocutionary)A(ct)P.” In Dayal’s analysis, questions in (1), (2), and (3) are of CP, SAP, and PerspP size, respectively. Essentially, in addition to a c-selectional constraint blocking the complementation of exceedingly large interrogative clauses like SAPs, which accounts for the ungrammaticality of (2), the proposal exploits a not-at-issue potential activeness (p-activeness) requirement introduced by PerspP, covering the pattern in (3). This proposal crucially provides a novel formal encoding of the division of labor among the multiple possible functions carried by the left periphery, offering revealing insights into where in the internal structure of a question the interrogative flavor enters.

The central goal of this paper is modest: we develop a point that Dayal hints at but does not elaborate on, namely that the three-way division of the left periphery is also systematically attested in Mandarin. To illustrate the three-way division of the Mandarin left periphery, the particle that plays a central role is *ma*, which marks polar questions. The syntactic, semantic, and pragmatic properties of *ma* have been extensively studied, often in connection with other question markers. To name several early references, see Chao 1926, 1965, Lü 1980, Hu 1981a,b, Li & Thompson 1981, Zhu 1982, Tang 1988, and Cheng 1991. A recent synthetic analysis is offered by Pan (2015, 2017, 2018, 2019, 2022), who argues that *ma* occupies a peripheral projection dubbed *illocutionary force phrase* (iForceP), which hosts two other question markers: *meiyóu* and *ba*. Like *ma*, *meiyóu* marks polar questions, but it additionally conveys perfective aspect with or without the perfective suffix *-le*.

Ba, on the other hand, marks confirmation questions, often translated into English as tag questions. Alongside the declarative base in (4), examples of these three question markers are presented in (5–7).

- | | |
|---|---|
| <p>(4) Zuotian xia-le yu.
yesterday fall-PFV rain
'It rained yesterday.'</p> | <p>(5) Zuotian xia(-le) yu meiyou?
yesterday fall-PFV rain MEIYOU
'Did it rain yesterday?'</p> |
| <p>(6) Zuotian xia-le yu ma?
yesterday fall-PFV rain MA
'Did it rain yesterday?'</p> | <p>(7) Zuotian xia-le yu ba?
yesterday fall-PFV rain BA
'It rained yesterday, didn't it?'</p> |

Although previous analyses that group these question markers within a single syntactic projection seem intuitive in light of their complementary distribution (e.g., Hu 1981a, Zhu 1982, Paul 2014, 2015, Pan 2015, 2017, 2018, 2019, 2022, Pan & Paul 2016, Paul & Pan 2016; cf. B. Li 2006, Kim 2019), we argue instead that their non-co-occurrence stems from a semantic clash rather than solely from a syntactic competition, and each of the question markers occupies a distinct projection: "ForceP (for *meiyou*) < PerspP (for *ma*) < Commit(ment)P (for *ba*)," reflecting Dayal's "CP < PerspP < SAP" division. Our primary evidence comes from their embeddability profiles, where the prior discussion has largely converged on the generalization that *meiyou* is consistently embeddable, while *ma* and *ba* are not. Drawing on both published data and collected judgments from 12 anonymous survey participants under experimental conditions, further corroborated by 14 additional native speakers of Mandarin, we refine this picture by showing that *ma* is in fact embeddable, albeit under restricted conditions. This results in a trichotomy in embeddability profiles of Mandarin question markers as stated in (8).

- (8) a. *Meiyou* can be embedded by question-embedding predicates.
b. *Ma* can be embedded by question-embedding predicates under restricted conditions.
c. *Ba* cannot be embedded by question-embedding predicates.

We extend this trichotomy to other Mandarin question markers insofar as they qualify as such, which we mean to include the *wh*-operator, A-not-A, *bu*, *ne*, and rising declaratives. The empirical contribution of this paper is thus to establish a novel data paradigm for the embeddability of Mandarin interrogative clauses, bringing to light a previously unnoticed third possibility.

Building on Dayal's proposal of a three-way division in the left periphery — particularly the middle projection identified as PerspP, which introduces the p-activeness requirement — we account for the trichotomy in embeddability profiles of question markers by positing a joint contribution of syntax and semantics to the selection of complements. Syntactically, c-selectional constraints determine possible embedded clause sizes (or types), whereas semantically, PerspP imposes the p-activeness requirement. Question-embedding predicates may select a ForceP or a PerspP, but they cannot embed CommitP, the highest projection among the three. Among embeddable interrogative clauses, the p-activeness requirement on the PerspP-level question markers checks whether the interpretation of the sentence is compatible with a context in which the question is p-active for the perspectival center, thereby regulating its embeddability, a constraint absent for ForceP question markers like *meiyou*. In this way, our proposal captures the observed trichotomy and correctly predicts that the embeddability of *ma* depends on two conditions: (i) the embedding context must license the appropriate clause size (or type), and (ii) the p-activeness requirement must be met. We summarize the central claims of our proposal in (9).

- (9) a. The iForceP has a more fine-grained internal structure: ForceP < PerspP < CommitP.
 b. The non-co-occurrence of the question markers in different projections is due to a semantic clash.
 c. The question markers in the PerspP introduce the p-activeness requirement.

Beyond the three-way division in the left periphery, we demonstrate that the empirical landscape is best accounted for under two theoretical commitments. First, an assertion-question distinction and a monopolar-bipolar distinction (cf. Biezma & Rawlins 2012, Krifka 2015, 2022, Rudin 2022, Matthewson 2025) are present in the language and necessary for the theory. Question markers differ in whether they yield a proposition, a singleton set of propositions, or a plural set of propositions. Our evidence comes from documented assertiveness diagnostics, as well as contrasts in the compatibility of question markers with contexts controlled for alternative saliency and with the attitude particle, the bias marker, and ‘right’ as a response.

Second, contra Dayal’s view that clause-typing uniformly occurs in a single projection (CP), we show that a clause can become a question at multiple syntactic levels. Presenting further evidence from a range of not necessarily genealogically related languages, we highlight this additional and crucial dimension of crosslinguistic variation concerning the locus of clause-typing in the left periphery, contributing a new perspective to the long-standing debate over where a clause acquires the interrogative flavor. In this spirit, we depart from Dayal’s perspective of a one-to-one correspondence between semantic transformations and syntactic projections in the interrogative left periphery, proposing instead that semantic transformations are mapped to a *minimal* size of syntactic projections. Together, these consequences further our understanding of how interrogative structure is distributed across the left periphery and how semantic distinctions interact with the syntactic architecture.

The rest of this paper is structured as follows. Section 2 reviews the traditional view on the cartography and embeddability of Mandarin question markers. Section 3 refines previous generalizations and arrives at a trichotomy in the embeddability profiles of these question markers. Section 4 presents a full analysis of their embeddability profiles, explicating how Dayal’s three-way split in the left periphery is systematically manifested in Mandarin and offering a semantic account for their non-co-occurrence. Section 5 fully fleshes out the semantic account by establishing the assertion-question distinction and the monopolar-bipolar distinction. Section 6 presents a comparison between Hindi/Urdu and Mandarin, highlighting a difference in terms of where clause-typing takes place. Section 7 concludes.

2 Traditional view

Before we move on to review prior analyses, we briefly touch upon the debates that have long centered on whether certain question markers under discussion exhibit functions beyond question marking. The relevant data are provided in (10–11), where we use subscripts to indicate distinct usages of a phonologically identical sentence-final particle (SFP) without necessarily implying a theoretical commitment as to whether these represent the same or different lexical items.¹

¹ Glossing abbreviations follow the Leipzig Glossing Rules with these additions: EXP = experiential, PQP = polar question particle, SUB = subordinator.

- (10) a. Ta bei she yao-guo **ma₂**. *declarative + ma₂*
 3SG PASS snake bite-EXP MA₂
 ‘You see, he has been bitten by a snake.’ (Pan 2019: 69)
- b. Jinlai **ma₂**! *imperative + ma₂*
 Come.in MA₂
 ‘Just come in!’ (Kim 2019: 19, adapted)
- c. Ni zai shuo shenme **ma₂**? *interrogative + ma₂*
 2SG PROG say what MA₂
 ‘Just tell me, what are you saying?’ (Kim 2019: 24, adapted)
- (11) a. Ta zuotian yinggai lai-guo **ba₂**. *declarative + ba₂*
 3SG yesterday should come-EXP BA₂
 ‘Probably, she came yesterday.’ (Pan 2019: 69)
- b. Ni xiang-yi-xiang **ba₃**! *imperative + ba₃*
 2SG think-one-think BA₃
 ‘Why don’t you think about it a little?’ (Li & Thompson 1981: 308)
- c. Ni daodi yao shenme **ba₄**? *interrogative + ba₄*
 2SG on.earth want what BA₄
 ‘Just tell me, what on earth do you want?’ (Chao 1965: 807)

Ma₂ “expresses that the reason is obvious” (Ding et al. 1961: 215; cf. Chappell 1991), “assumes that the other person doesn’t know” (Zhu 1982: 213), or operates as “a speech act intensifier” (Kim 2019: 5), while *ba₂* “expresses a speculative tone, somewhere between doubt and belief” (Ding et al. 1961: 213), *ba₃* is used for “requesting, advising, urging, and commanding” (Ding et al. 1961: 211), and *ba₄* occurs “in impatience scenarios” (Ettinger & Malamud 2019: 245) and carries “an ‘unfriendly’ effect” (X. Yuan 2020: 470). Chao (1965) and Zhu (1982) treat *ba₃* and *ba₄* consistently as an “advisative particle” (Chao 1965: 809) or “imperative marker” (Zhu 1982: 211), noting that the imperative force lies in prompting the addressee to provide an answer when *ba₃₋₄* is attached to an interrogative. Alternative analyses that treat *ba₂₋₄* and the confirmation question marker *ba* uniformly suggest that they as one lexical item convey “the uncertain mood” (Hu 1981b: 416, cf. Chu 1998, 2009, Chu & Li 2004), have “the effect of soliciting the approval or agreement of the hearer” (Li & Thompson 1981: 307), or “weaken the speaker’s commitment (i.e., neustic)” (Han 1995: 100, cf. B. Li 2006, Kim 2019). See also Yang 2021 for a unified analysis of *ba₂* and *ba₄* and X. Yuan 2021 for a unified analysis of *ba* and *ba₂*.² Notably, while *ba* and *ba₂₋₄* are usually written with the same character, *ma₂* can be written in multiple ways, all of which use a different character from the polar question marker *ma* and potentially correspond to phonologically distinct but similar forms, such as *me* as documented by Chao (1926, 1965), Ding et al. (1961), and Chappell (1991) and *mo* as documented by M.-T. Liu (1964), Hu (1981b), and Zhu (1982).

² X. Yuan (2021) further notes that the interpretation of *ba/ba₂* depends on whether the accompanying intonation is falling or rising. It is therefore important to clarify that all examples cited or provided in this paper concern cases with falling intonation, even when a question mark is used to highlight the confirmation-seeking use of certain examples. In the absence of experimental evidence clarifying the relationship between the intonation found in rising *balba₂* utterances and that in rising declaratives or that typically associated with information-seeking polar questions (e.g., *ma* questions), we leave rising *balba₂* utterances for future work.

Although detailed analyses of ma_2 and ba_{2-4} lie beyond the scope of this paper, as discussed later, ultimately, our proposal adopts a stance on a part of these debates. Our arguments favor the position that ma_2 and the polar question marker ma are distinct lexical items, and ba_2 and the confirmation question marker ba are the same lexical item, while remaining agnostic about the status of ba_{3-4} . For now, we set aside these debates and focus only on the polar question use of ma and the confirmation question use of ba , which we believe are more directly revealing of the central issue at hand.

2.1 Cartography of the left periphery

The traditional view of the syntactic statuses of *meiyou*, *ma*, and *ba* has not entertained the possibility that they each belong to their own projection, and a key observation motivating this view is that they never co-occur with each other, regardless of order. Examples are provided in (12–14).

(12) *Non-co-occurrence of meiyou and ma*

Intended: ‘Did it rain yesterday?’

a. *Zuotian xia-le yu **meiyou ma**?
yesterday fall-PFV rain MEIYOU MA

b. *Zuotian xia-le yu **ma meiyou**?
yesterday fall-PFV rain MA MEIYOU

(13) *Non-co-occurrence of meiyou and ba*

Intended: ‘Did it rain yesterday?’/‘It rained yesterday, didn’t it?’

a. *Zuotian xia-le yu **meiyou ba**?
yesterday fall-PFV rain MEIYOU BA

b. *Zuotian xia-le yu **ba meiyou**?
yesterday fall-PFV rain BA MEIYOU

(14) *Non-co-occurrence of ma and ba*

Intended: ‘Did it rain yesterday?’/‘It rained yesterday, didn’t it?’

a. *Zuotian xia-le yu **ma ba**?
yesterday fall-PFV rain MA BA

b. *Zuotian xia-le yu **ba ma**?
yesterday fall-PFV rain BA MA

Note again that (12a) should not be confused with a segmentally identical string where *ma* is replaced with ma_2 , which is acceptable under a different interpretation as in (15).

(15) Zuotian xia-le yu **meiyou ma_2** ?

yesterday fall-PFV rain MEIYOU MA_2

‘Just tell me, did it rain yesterday?’

Owing to the observed non-co-occurrence of question markers, prior studies have generally treated them as belonging to the same category and realizing the same functional head.

The discussion on the cartography of SFPs dates back to Hu (1981a) and Zhu (1982), who propose a tripartite classification of SFPs. They further observe that particles from different categories may co-occur but only in a fixed linear order as in (16), where we underline the category into which the question markers under discussion are classified.

(16) *Hu 1981a and Zhu 1982*

- a. structural/tense particles < consonantal particles < vocalic particles (Hu 1981a: 348)
- b. tense particles < imperative/interrogative particles < attitude/emotion particles (Zhu 1982: 207–214, adapted)

In their generalizations, since both *ma* and *ba* have consonantal onsets and convey imperative or interrogative force, they are classified as consonantal particles or imperative/interrogative particles and thus occupy the same position in the linear sequence. *Meiyoud*, which is typically not considered a particle potentially due to its lexical semantic interpretation (composed of the perfective negation morpheme *mei* and the perfective marker *you*, as discussed in Pan 2019 for example), is not included in their discussion.

The theoretical formulation of the Mandarin left periphery begins with Paul (2005), who demonstrates that it aligns only partially with the hierarchy proposed by Rizzi (1997, 2004) and Belletti (2004). Inspired by Hu and Zhu, Paul (2014, 2015), Pan & Paul (2016), and Paul & Pan (2016) extend previous analyses of *ma* as a complementizer (e.g., H.-T. T. Lee 1986, Cheng 1991) and recast the tripartite classification and the fixed linear order as a split CP that has three layers as in (17).

(17) *Paul 2014, 2015, Pan & Paul 2016, and Paul & Pan 2016*

- LowC(omplementizer)P < ForceP < Att(itude)P (Paul & Pan 2016: 56–68, adapted)

Under this proposal, both *ma* and *ba* function as “heads expressing the sentence type” (Paul 2014: 90) and are therefore analyzed as force complementizers, while the status of *meiyoud* is not discussed. In a similar vein, Erlewine (2017) proposes three classes of SFPs, with *ma* classified as a clause-typing SFP, positioned above low SFPs (argued to sit at the lower phase edge) and below attitude SFPs.

Building upon the work on the split Mandarin CP, Pan (2015, 2017, 2018, 2019, 2022) incorporates a substantially broader range of sentence-final elements and uses the co-occurrence ordering test to construct a more fine-grained representation of the Mandarin left periphery as in (18).

(18) *Pan 2015 et seq.*

- S(entential)Asp(ect)P < OnlyP < i(llocutionary)ForceP < S(pecial)Q(uestion)P < Att(itude)P (Pan 2022: 122, adapted)

Pan proposes a division of ForceP into iForceP and SQP. Since *meiyoud*, *ma*, and *ba* are all “used to indicate the illocutionary force” (Pan 2015: 837), and they do not express “something else than a request for information” (Pan 2015: 843) in an interrogative form, Pan places them in the iForceP.

Importantly, to our knowledge, no previous study has proposed splitting *meiyoud*, *ma*, and *ba* into distinct projections, a configuration that we ultimately argue for. Against this background, the next subsection reviews previous observations and analyses concerning the (un)embeddability of these question markers, recognizing it, in line with prior work, as the most crucial diagnostic for their cartographic status.

2.2 Dichotomy in question marker embeddability

Previous work (e.g., Li & Thompson 1981, Paul 2014, Pan 2015) observes that *meiyoud* can always embed, while *ma* and *ba* never embed. This observation is illustrated in (19), where the question markers are embedded in the context of a noun complement or a clausal subject. Examples from the literature in which question markers are embedded within clausal objects are presented in (20).

(19) *Noun complement and clausal subject*

[Zuotian xia-le yu **meiyou**/***ma**/***ba**] (de wenti) hen guanjian.

Yesterday fall-PFV rain MEIYOU MA BA COMP question very crucial

(Intended:) ‘(The question of) whether it rained yesterday is very crucial.’

(20) *Clausal object*

a. Wo bu zhidao [ni chi-le fan **meiyou**].

1SG not know 2SG eat-PFV meal MEIYOU

‘I don’t know if you ate.’

(Pan 2015: 841)

b. *Wo bu (xiang) wen ni [ta mingtian hui lai **ma**?]

1SG not want ask 2SG 3SG tomorrow will come MA

Intended: ‘I don’t (want to) ask you whether you will come.’

(Tang 1988: 297)

c. *Wo xiang zhidao [ni shi Xibanya ren **ba**].

1SG want know 2SG COP Spain person BA

Intended: ‘I want to know whether you are Spanish.’

(Pan 2015: 842)

The observed embeddability difference is accounted for by the lexical properties of question markers. For instance, Paul (2014) proposes that SFPs carry a $[\pm\text{root}]$ feature, and only $[-\text{root}]$ SFPs are embeddable. Under this proposal, *ma* and *ba* are $[\text{+root}]$, and while Paul does not discuss *meiyou*, given its embeddability, it would be considered $[-\text{root}]$. The interpretability of the $[\pm\text{root}]$ feature, however, is not addressed.

This lexical account would be called into question if *ma* can be shown to be embeddable after all, and establishing this is one of the empirical aims of the present paper. We first draw attention to an example from the literature that has not received much discussion to our knowledge. Consider (21).

(21) Wo wen ni [ta mingtian hui lai **ma**?]

1SG ask 2SG 3SG tomorrow will come MA

‘I ask you whether he will come tomorrow.’

(Tang 1988: 295)

Tang (1988) punctuates the sentence in (21) (and likewise back in (20b)) with a question mark, even though the main clause is declarative. This may reflect the fact that the subordinate clause is most naturally produced with the intonation pattern typical of Mandarin information-seeking *ma* questions. To avoid confusion, in what follows, we punctuate examples like (21) as in (22): the sentence ends with a full stop, but the interrogative intonation is indicated by an upward arrow.³

(22) Wo wen ni [ta mingtian hui lai **ma**]_↑.

1SG ask 2SG 3SG tomorrow will come MA

‘I ask you whether he will come tomorrow.’

(Tang 1988: 295, adapted)

Although Tang analyzes (21) as an instance of “direct discourse,” which remains a possible interpretation of this specific example, we demonstrate in the next section that newly available data and diagnostics compel us to

³ We use “↑” in accordance with the convention in the literature, although we realize that the intonation associated with *ma* questions is not necessarily rising. See Liu & Xu 2005 for a detailed discussion of Mandarin question intonation. Note, however, that their terminology differs from ours: in particular, what they term *yes-no questions* correspond to what we call *rising declaratives*, while *ma* questions fall under their category of *particle questions*.

reconsider *ma* as indeed embeddable. This is where the third possibility of a question marker's embeddability becomes relevant, motivating a new analysis that supersedes the lexical account.

3 New perspective

Another example that may challenge the lexical account's prediction of the embeddability of *ma* questions comes from Bhatt & Dayal (2020). We copy their example in (23).⁴

- (23) John xiang zhidao [xia yu le **ma** zuotian]_↑.
 John want know fall rain PRF MA yesterday
 'John wants to know whether it rained yesterday.' (Bhatt & Dayal 2020: 1122)

Prompted by Bhatt & Dayal, we add in (24) examples involving pronoun coreference and variable binding, which are evidence that the questions are not quotations and are truly embedded. If they were quotations, personal pronouns would behave differently (McCloskey 2006) as in (25).

- (24) *Embedded questions: 1sg = speaker and 3sg = matrix subject*
- a. Xiaohong_i xiang zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaohong want know 3SG POSS score than 1SG high MA
 'Xiaohong_i wants to know whether her_i score is higher than mine_j.'
 - b. [Mei ge ren]_i dou xiang zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 every CLF person all want know 3SG POSS score than 1SG high MA
 'Everyone_i wants to know whether their_i score is higher than mine_j.'
- (25) *Quoted questions: 1sg = quoting predicate subject and 3sg = someone else*
- a. Xiaohong_i xiang zhidao: ["ta_j de fenshu bi wo_i gao **ma**?"]
 Xiaohong want know 3SG POSS score than 1SG high MA
 'Xiaohong_i wants to know, "is her_j score higher than mine_i?"'
 - b. [Mei ge ren]_i dou xiang zhidao: ["ta_j de fenshu bi wo_i gao **ma**?"]
 every CLF person all want know 3SG POSS score than 1SG high MA
 'Everyone_i wants to know, "is their_j score higher than mine_i?"'

An example of contexts in which utterances like that in (24a) can be felicitously made is one in which the speaker acts as a proxy for the matrix subject, posing a question on their behalf and declaring their curiosity about the answer, such as (26).

⁴ Following Bhatt & Dayal's (2020) analysis, we bracket the example as if 'yesterday' forms part of the embedded question, but we acknowledge the possibility that the position of 'yesterday' may result from right dislocation (e.g., Chao 1965, Cheung 2009, T. T.-M. Lee 2017, Yip 2025). It could, in fact, occupy a position in the matrix periphery. Nevertheless, neither their nor our argument depends on the precise analysis of the position of 'yesterday', as the sentence remains acceptable and retains the intended interpretation when 'yesterday' appears in its default position:

- (i) John xiang zhidao [zuotian xia yu le **ma**]_↑.
 John want know yesterday fall rain PRF MA
 'John wants to know whether it rained yesterday.'

(26) *Context for (24a)*

(Xiaohong and John both took a seminar by Mary. The grades had not been posted yet, and Xiaohong wanted to know whether she or John got a higher grade. Being a very shy student, she did not want to ask Mary herself. She asked John to ask Mary. So, John said to Mary, “Xiaohong wants to know whether her score is higher than mine.”)

Data resembling (24) have been confirmed in Beijing Mandarin under experimental settings by Y. Liu (2025) using a novel semantic judgment elicitation paradigm.⁵ While we believe that the acceptability is found in a much wider range of dialects of Mandarin—including those spoken in Fuzhou, Hanzhong, Hong Kong, Mianyang, Nanning, Wuhan, and Xi’an, whose speakers we have consulted—we acknowledge regional variations, which we leave for future work.⁶

3.1 Trichotomy in question marker embeddability

The embeddability of *ma* depends heavily on the matrix predicate type. While predicates like ‘want to know’ accept it, other predicates like ‘know’ reject it as in (27).

- (27) #Xiaohong_i zhidao [ta_i de fenshu bi wo_j gao **ma**]_(↑).
 Xiaohong know 3SG POSS score than 1SG high MA
 Intended: ‘Xiaohong_i knows whether her_i score is higher than mine_j.’

⁵ To briefly summarize the experimental setup, Y. Liu’s (2025) novel semantic judgment elicitation paradigm presented participants with the following instruction:

- (i) Below are some pairs of expressions, each with an accompanying context. In each pair of expressions, option (a) consists of two sentences, and option (b) attempts to convey the same meaning using only one sentence. Your task is to understand the accompanying context, read both expressions aloud, and decide whether the two sentences in (a) can be naturally conveyed using the single sentence in (b) in the accompanying context. If you believe it is possible, choose (b). If not, choose (a). If both (a) and (b) seem unacceptable for you, choose (a) as well. (Y. Liu 2025: 94)

For each item under each condition, the design essentially forced participants to choose between the target sentence and a two-sentence paraphrase of it, in which the first sentence conveyed the matrix content, and the second conveyed the embedded content. To illustrate, the two-sentence paraphrase of (24a) in the experiment is given below:

- (ii) Xiaohong_i xiang zhidao yi jian shi. Na jiu shi [ta_i de fenshu shi-bu-shi bi wo_j gao].
 Xiaohong want know one CLF thing that just COP 3SG POSS score COP-not-COP than 1SG high
 ‘Xiaohong_i wants to know one thing. That thing is whether her_i score is higher than mine_j.’ (Y. Liu 2025: 94)

Subscripts here are only for clarity and were not included in the survey. Pronoun referents were made explicit in the accompanying context. The results revealed a significant contrast in the acceptability of embedding *ma* questions between quasi subordinators and non-quasi subordinators, as well as between matrix declaratives and matrix interrogatives. We discuss these contrasts in Section 3.1.

⁶ As an illustration of regional variations, Taiwanese speakers may find embedded *ma* questions much less acceptable even in contexts where we predict them to be possible, as reported by Pan (2015, 2019) a.o.:

- (i) a. *Ta wen wo [ni xihuan Faguo ma].
 3SG ask 1SG 2SG like France MA
 Intended: ‘He asks me whether you like France.’ (Pan 2015: 840)
 b. *Ta xiang zhidao [ni qu-guo Faguo ma].
 3SG want know 2SG GO-EXP France MA
 Intended: ‘He wants to know whether you have been to France.’ (Pan 2019: 100)

Both examples cited here are acceptable in Beijing Mandarin.

We have also consulted Mandarin speakers who are also native to the Jiaxing or Shanghainese dialect of Wu Chinese, and they find *ma* questions uniformly embeddable. We leave a more systematic investigation of these data points to future work.

Building on Woods (2016) and Dayal’s (2025) observations of English EIQs, we find that the observed contrast between ‘want to know’ and ‘know’ does not align with the familiar rogative-responsive distinction. This can be seen from predicates like ‘forget’, which is responsive but patterns with ‘want to know’ instead of ‘know’, as it allows the embedding of *ma* (28).⁷

- (28) Xiaohong_i wangji-le [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaohong forget-PFV 3SG POSS score than 1SG high MA
 ‘Xiaohong_i forgot whether her_i score is higher than mine_j.’

The nature of this contrast has to do with whether the predicate is semantically compatible with a context in which the possibility that the matrix subject does not know the answer to the embedded question exists: *ma* necessitates the manifestation of this possibility in the context, a condition we refer to as *ignorance requirement* (IR). We follow Dayal & Grimshaw (2009) and term a predicate that satisfies the IR (e.g., ‘want to know’) a *quasi subordinator* (QS) and one that does not (e.g., ‘know’) a *non-quasi subordinator* (NQS). An illustration of the contrast between QS and NQS in terms of the IR as well as a comparison with rogative and responsive predicates is presented in Table 1.⁸

Table 1 Question embedding predicates of different types

	QS (✓IR): ◇¬know(<i>x</i> , answer(<i>Q</i>))	NQS (✗IR): □know(<i>x</i> , answer(<i>Q</i>))
Rogative: [+Q]	<i>ask, want to know, wonder</i>	
Responsive: [±Q]	<i>forget</i>	<i>explain, know, remember</i>

Additional support for the IR arises from the way an NQS exhibits shiftiness under matrix modality, negation, and interrogative. Although the degree of acceptability shows minor variation across speakers for some of the examples, all consultants agree that (29–31) present a clear contrast with (27).⁹

(29) *Modality*

- a. (?)Huoxu/yexu Xiaohong_i zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 maybe maybe Xiaohong know 3SG POSS score than 1SG high MA
 ‘Maybe Xiaohong_i knows whether her_i score is higher than mine_j.’
- b. (?)Xiaohong keneng zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaohong may know 3SG POSS score than 1SG high MA
 ‘Xiaohong_i may know whether her_i score is higher than mine_j.’

⁷ Some consultants find the sentence unacceptable when it’s uttered out of the blue. When proper contexts are provided, however, the sentence feels very natural. We provide an example of the contexts:

(i) Context for (28)

Xiaohong is in charge of noting the grades of the whole class, but she is very forgetful. One day, she finds that she has forgotten my grade for the last exam, and asks me if my grade is higher than hers. Being a very forgetful person as well, I decide to ask Xiaoming, who knows every details of everything. So I say to Xiaoming: “Xiaohong forgot, is her score higher than mine?”

⁸ For simplicity, we treat ‘want to know’ as a single unit here, but Section 4.2 will discuss ‘want to know’ in compositional terms.

⁹ In (31), the matrix interrogative makes use of another strategy for forming polar questions in Mandarin — namely, the A-not-A template. Section 3.2 discusses its embeddability profile in detail.

(30) *Negation*

Xiaohong bu zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaohong not know 3SG POSS score than 1SG high MA
 ‘Xiaohong_i doesn’t know whether her_i score is higher than mine_j.’

(31) *Interrogative*

Xiaohong_i zhi-bu-zhidao [ta_i de fenshu bi wo_j gao **ma**]_↑?
 Xiaohong know-not-know 3SG POSS score than 1SG high MA
 ‘Does Xiaohong_i know whether her_i score is higher than mine_j?’

Even though the matrix predicate is an NQS and cannot satisfy the IR by itself, the presence of matrix modality, negation, or interrogative opens up the possibility that the matrix subject is ignorant of the answer to the embedded question, thereby satisfying the IR. Among these conditions, the effect of the matrix interrogative has been confirmed under experimental settings by Y. Liu (2025). See McCloskey 2006 and Woods 2016 for a more detailed discussion of these conditions in connection with English EIQs.

The embeddability profile we identify for *ma* appears to be new to the Chinese linguistics literature, especially in light of its semantic nature. This motivates a reassessment of the embeddability of other question markers, including *meiyou* and *ba*. Unlike *ma*, their (un)embeddability remains unaffected regardless of whether the IR is satisfied and aligns with the traditional view: *meiyou* questions (32) are always embeddable, while *ba* questions (33) never are (although *ba* can occur in quotations).¹⁰

- (32) a. Xiaohong_i xiang zhidao [ta_i jige-le **meiyou**]. QS + meiyou question
 Xiaohong want know 3SG PASS-PFV MEIYOU
 ‘Xiaohong wants to know if he passed.’
 b. Xiaohong_i zhidao [ta_i jige-le **meiyou**]. NQS + meiyou question
 Xiaohong know 3SG PASS-PFV MEIYOU
 ‘Xiaohong knows if he passed.’

¹⁰ The perfective aspect of *meiyou* makes ‘whether his score is higher than mine’ sound unnatural with it. While (32), as a result, fails to maintain a minimal pair relation with the *ma* question examples presented thus far, the embeddability profile of *ma* questions can be replicated using ‘whether he passed’:

- (i) a. Xiaohong_i xiang zhidao [ta_i jige-le **ma**]_↑. QS + ma question
 Xiaohong want know 3SG PASS-PFV MA
 ‘Xiaohong wants to know whether he passed.’
 b. #Xiaohong_i zhidao [ta_i jige-le **ma**]_↑. #NQS + ma question
 Xiaohong know 3SG PASS-PFV MA
 Intended: ‘Xiaohong knows whether he passed.’

In the remainder of the paper, particularly when we turn to question markers beyond *meiyou*, *ma*, and *ba*, it should be noted that some other question markers may also exhibit some degree of idiosyncrasy as *meiyou* does here. Consequently, no single declarative base can serve uniformly across all question markers. Nonetheless, we have endeavored to keep the relevant examples as minimally different from each other as possible.

- (33) a. *Xiaohong_i xiang zhidao [ta_i de fenshu bi wo_j gao **ba**]_(↑). *QS + ba question
 Xiaohong want know 3SG POSS score than 1SG high BA
 Intended: ‘Xiaohong_i wants to know whether her_i score is higher than mine_j.’
- b. *Xiaohong_i zhidao [ta_i de fenshu bi wo_j gao **ba**]_(↑). *NQS + ba question
 Xiaohong know 3SG POSS score than 1SG high BA
 Intended: ‘Xiaohong_i knows whether her_i score is higher than mine_j.’

We thus observe a trichotomy in the (un)embeddability of question markers in clausal objects. Together with the previous observations concerning noun complements and clausal subjects, we summarize the embeddability profiles of *meiyou*, *ma*, and *ba* in Table 2.¹¹

Table 2 Embeddability profiles of *meiyou*, *ma*, and *ba*

	<i>Meiyou</i>	<i>Ma</i>	<i>Ba</i>
Noun complement	✓	✗	✗
Clausal subject	✓	✗	✗
Clausal object (IR not satisfied)	✓	✗	✗
Clausal object (IR satisfied)	✓	✓	✗

In what follows, we extend the observed trichotomy to other Mandarin question markers (to the extent that they can plausibly be considered question markers). We examine the embeddability profiles of other question markers identified by Paul 2014 et seq. in ForceP and by Pan 2015 et seq. in iForceP, including the *wh*-operator, *bu*, and *ne*. Another commonly used polar question-forming strategy, A-not-A, would also merit a reassessment of its embeddability. We then conclude the section with a discussion of rising declaratives and a comment on the difference in embeddability between *ma* and *ma*₂.

3.2 Question markers with *meiyou*-type embeddability

Mandarin *wh*-questions are embeddable in all question embedding contexts (e.g., Huang 1982a,b, Cheng 1991) as in (34). They pattern like *meiyou* in terms of embeddability.

- (34) a. *Noun complement or clausal subject*
 [Mingtian **shenme shihou** xia yu] (de wenti) hen guanjian.
 tomorrow what time fall rain COMP question very crucial
 ‘(The question of) when it will rain tomorrow is very crucial.’
- b. *Clausal object*
 Xiaohong_i (xiang) zhidao [ta_i de fenshu **weishenme** bi wo_j gao].
 Xiaohong want know 3SG POSS score why than 1SG high
 ‘Xiaohong_i knows/wants to know why her_i score is higher than mine_j.’

¹¹ The DP shell analysis (Lees 1960, Rosenbaum 1965, Davies & Dubinsky 1998, Takahashi 2010, Hartman 2012) claims that clausal subjects are DPs, formed by phonologically null D heads that embed CPs. Under this analysis, clausal subjects may be understood to form a subset of noun complements. Our proposal does not yield any predictions that conflict with the DP shell analysis, as we find no instances in which a question marker’s embeddability differs between noun complements and clausal subjects.

A-not-A signals polar questions. It is formed by reduplicating the verb or its leftmost portion and placing a negative morpheme in between (e.g., Li & Thompson 1979, Huang 1982b, 1991, Zhu 1985) as in (35).¹²

- (35) Mingtian **xia-bu-xia** yu?
 tomorrow fall-not-fall rain
 ‘Will it rain tomorrow?’

As shown in (36), A-not-A questions are also embeddable across the board, resembling *meiyou* questions.

- (36) a. *Noun complement or clausal subject*
 [Mingtian **xia-bu-xia** yu] (de wenti) hen guanjian.
 tomorrow fall-not-fall rain COMP question very crucial
 ‘(The question of) whether it will rain tomorrow is very crucial.’
 b. *Clausal object*
 Xiaohong_i (xiang) zhidao [ta_i **ji-mei-jige**].
 Xiaohong want know he pass-not.PFV-pass
 ‘Xiaohong_i knows/wants to know whether she_i passed.’

Following previous analyses, we assume that both *wh*- (e.g., Y.-H. A. Li 1992, Aoun & Li 1993a,b, Tsai 1994, Soh 2005) and A-not-A (e.g., Ernst 1994, Schaffar & Chen 2001, Law 2006, Tsai & Yang to appear) questions involve a dependency with a functional head or an operator bearing some interrogative clause-typing feature (e.g., [+Q]) in the left periphery. Any subsequent claims regarding the cartography of *wh*- or A-not-A questions are directed toward this left peripheral element that bears the clause-typing feature.

3.3 Question markers with *ma*-type embeddability

The negation morpheme *bu* ‘not’ can be used as an SFP and signal polar questions (Pan 2019) as in (37).

- (37) Mingtian xia yu **bu**?
 tomorrow fall rain BU
 ‘Will it rain tomorrow?’

We observe that *bu* questions exhibit the same embeddability pattern as *ma* questions, as presented in (38). They do not embed under noun complements or clausal subjects, but they may embed under clausal objects only when the IR is satisfied.

¹² The literature further distinguishes variants of A-not-A where the reduplicated morpheme is a modal (e.g., *hui* ‘will’) or a focus marker (lexically identical to the copula *shi*), often referred to as *M-not-M* and *B-not-B* (e.g., Hagstrom 2006), respectively. B-not-B is further set apart from the others as *outer A-not-A* in contrast to *inner A-not-A* (e.g., Tsai & Yang to appear). These variants do not differ in their embeddability profiles, and we discuss A-not-A in its strictest sense, excluding constructions that superficially resemble A-not-A but have been argued not to involve reduplication (e.g., VP-not-V) and referring only to the inner type.

(38) a. *Noun complement or clausal subject*

*[Mingtian xia yu **bu**]_(↑) (de wenti) hen guanjian.
 tomorrow fall rain BU COMP question very crucial
 Intended: ‘(The question of) whether it will rain tomorrow is very crucial.’

b. *Clausal object (IR not satisfied)*

#Xiaohong_i zhidao [ta_i de fenshu bi wo_j gao **bu**]_(↑).
 Xiaohong know 3SG POSS score than 1SG high BU
 Intended: ‘Xiaohong_i knows whether her_i score is higher than mine_j.’

c. *Clausal object (IR satisfied)*

Xiaohong_i xiang zhidao [ta_i de fenshu bi wo_j gao **bu**]_(↑).
 Xiaohong want know 3SG POSS score than 1SG high BU
 ‘Xiaohong_i wants to know whether her_i score is higher than mine_j.’

Ne can be attached to interrogatives such as *wh*-, A-not-A, and *meiyou* questions, but not to *bu*, *ma*, or *ba* questions.¹³ In general, it signals a “follow-up question” (Paul 2014: 83) or “marks that the current discourse move is ‘unexpected’” (X. Yuan 2023: 748; cf. Chu 2009). When the interrogative base is a polar question, *ne* can give “a meaning along the lines of an English *or not* question” (Constant 2014: 339). An example of its usage with *meiyou* is in (39).

(39) Zuotian xia-le yu meiyou **ne**?

yesterday fall-PFV rain MEIYOU NE
 ‘Did it rain yesterday or not?’

The embeddability profile of *ne* resembles that of *bu* and *ma*. It is embeddable in a clausal object when the IR is satisfied but unembeddable elsewhere as in (40).

(40) a. *Noun complement or clausal subject*

*[Zuotian xia-le yu meiyou **ne**]_(↑) (de wenti) hen guanjian.
 yesterday fall-PFV rain MEIYOU NE COMP question very crucial
 Intended: ‘(The question of) whether it rained yesterday or not is very crucial.’

b. *Clausal object (IR not satisfied)*

#Xiaohong_i zhidao [ta_i jige-le meiyou **ne**]_(↑).
 Xiaohong know 3SG pass-PFV MEIYOU NE
 Intended: ‘Xiaohong_i knows whether she_i passed or not.’

c. *Clausal object (IR satisfied)*

Xiaohong_i xiang zhidao [ta_i jige-le meiyou **ne**]_(↑).
 Xiaohong want know 3SG pass-PFV MEIYOU NE
 ‘Xiaohong_i wants to know whether she_i passed or not.’

¹³ Like *ma* and *ba*, *ne* wears many hats. When used sentence-finally, *ne*₂ “exaggerates the matter” (Ding et al. 1961: 215; cf. Lü 1980, 1982, Wu 2005), expresses an “interest in additional information” (Chao 1965: 805), or “calls the other person’s attention to a specific point in what the speaker is saying” (Hu 1981b: 417) in declaratives, while *ne*₃ conveys a continuous aspect (e.g., Chao 1965, Lü 1980, Zhu 1982). Neither causes any confound in (39–40) or the other *ne* examples in this paper, as *ne*₂ occurs only in declaratives (cf. Chu 2009, X. Yuan 2023 for a unified treatment of *ne* and *ne*₂), and *ne*₃ is positioned lower than ForceP or iForceP (e.g., Paul 2014, Pan 2015). Additionally, since *ne*₃’s meaning is incompatible with the perfective aspect, it cannot co-occur with *meiyou* or the verbal suffix *-le*.

3.4 Questions that do not embed

We adopt the arguments advanced by Pan & Paul (2016) and Luo (to appear) and maintain that none of the question markers discussed thus far are “optional” in the sense of Cheng 1991, Bailey 2010, 2013, and Biberauer, Holmberg & Roberts (2014). Based on this commitment, rising declaratives (RDs) in Mandarin should be analyzed on a par with other question markers rather than as an intonational pattern that invariably accompanies interrogatives. Like in (2) for English, we use a double upward arrow to represent the intonation typically associated with RDs in Mandarin as well, as presented in (41).

- (41) [Mingtian xia yu?]_{↑↑}
 tomorrow fall rain
 ‘It will rain tomorrow?’

As in (42), similar to *ba* questions, rising declaratives do not embed.

- (42) a. *Noun complement or clausal subject*
 *[Mingtian xia yu]_{↑↑} (de wenti) hen guanjian.
 tomorrow fall rain COMP question very crucial
 Intended: ‘(The question of) whether it will rain tomorrow is very crucial.’
 b. *Clausal object*
 *Xiaohong_i (xiang) zhidao [ta_i de fenshu bi wo_j gao]_{↑↑}.
 Xiaohong want know 3SG POSS score than 1SG high
 Intended: ‘Xiaohong_i knows/wants to know whether her_i score is higher than mine_j.’

Another type of questions that do not embed in Mandarin is those followed by *ma*₂ (introduced in Section 2). The relevant data are presented in (43).

- (43) a. *Noun complement or clausal subject*
 *[Zuotian xia-le yu meiyou **ma**₂]_(↑) (de wenti) hen guanjian.
 yesterday fall-PFV rain MEIYOU **MA**₂ COMP question very crucial
 Intended: ‘(The question of) whether it rained yesterday is very crucial.’
 b. *Clausal object*
 *Xiaohong_i (xiang) zhidao [ta_i jige-le meiyou **ma**₂]_(↑).
 Xiaohong want know 3SG PASS-PFV MEIYOU **MA**₂
 Intended: ‘Xiaohong_i knows/wants to know whether she_i passed.’

The fact that *ma* and *ma*₂ differ in their embeddability supports the view that they are not a single lexical item, aligning with Pan’s (2019) classification of these particles as belonging to distinct projections (i.e., *ma* in iForceP and *ma*₂ in AttitudeP).¹⁴

¹⁴ Despite the embeddability evidence presented here in favor of treating *ma* and *ma*₂ as distinct particles, we acknowledge that our understanding of *ma*₂ remains limited, preventing us from offering a definitive account of its semantics. For example, while no prior work has explored the possibility that the occurrences of *ma*₂ following declarative, imperative, and interrogative clauses might represent distinct particles, whether such a treatment is accurate remains to be determined. Furthermore, empirically, we observe that *ma*₂ in interrogatives can co-occur with ForceP-level question markers but not with PerspP- or CommitP-level ones, but the source of this restriction — whether syntactic, semantic, or even phonological — remains unclear. As mentioned in Section 2, a detailed analysis of *ma*₂ lies beyond the scope of this paper, and we leave the investigation of its semantics and (non-)co-occurrence pattern for future work.

4 Conspiracy of syntax and semantics

The embeddability profiles covered thus far are summarized in Table 3. The observed trichotomy, at the very least, poses a serious empirical challenge to lexical analyses based on a binary feature such as $[\pm\text{root}]$, which by itself can account only for a two-way distinction. Alternative analyses that posit multiple features are also undermined by the semantic dimension inherent in the embeddability profile of *bu*, *ma*, and *ne*, not to mention the potential complications associated with the interpretability of the features. These considerations motivate a new analysis, one that integrates both syntactic and semantic components.

Table 3 Embeddability profiles of question markers

	<i>Wh-</i>	<i>Meiyou</i>	A-not-A	<i>Bu</i>	<i>Ma</i>	<i>Ne</i>	<i>Ba</i>	↑↑
Noun complement	✓	✓	✓	✗	✗	✗	✗	✗
Clausal subject	✓	✓	✓	✗	✗	✗	✗	✗
Clausal object (IR not satisfied)	✓	✓	✓	✗	✗	✗	✗	✗
Clausal object (IR satisfied)	✓	✓	✓	✓	✓	✓	✗	✗

4.1 Syntactic component: Split iForceP

We capture the tiered embeddability pattern using a three-way split iForceP and label the projections as force phrase (ForceP), perspective phrase (PerspP), and commitment phrase (CommitP). With respect to the grammatical functions and the nomenclature of the projections, we adopt ForceP in the sense of Rizzi 1997, 2001, 2004 (and Paul 2014, 2015 to some extent), PerspP in the sense of Dayal 2025, and CommitP in the sense of Miyagawa 2022 and Miyagawa & Hill 2025. As will become clearer in the remainder of this paper, ForceP marks the earliest stage of clause typing, PerspP triggers perspectival centering, and CommitP accommodates elements that modify the speaker’s commitment to the core proposition. In this respect, ForceP maps roughly onto Dayal’s CP, PerspP maps more closely onto Dayal’s PerspP, and CommitP, together with the projections above it, maps roughly onto Dayal’s whole SAP domain.

A neat correspondence emerges between the grammatical function of each of the three projections and its degree of embeddability. ForceP hosts the most embeddable elements, including the *wh*-operator, A-not-A, and *meiyou*. PerspP hosts elements embeddable under restricted conditions, including *bu*, *ma*, and *ne*. CommitP hosts the unembeddable element *ba*, and we contend that the realization of a rising declarative requires the size of CommitP. The revised periphery looks like (44).

(44) SAspP < OnlyP < ForceP < PerspP < CommitP < SQP < AttitudeP

- a. ForceP: *wh-*, *meiyou*, and A-not-A
- b. PerspP: *bu*, *ma*, and *ne*
- c. CommitP: *ba* and ↑↑

Recall that most of the aforementioned question markers cannot co-occur with each other, as presented in Table 4.¹⁵ Since our split iForceP proposal eliminates the syntactic competition between *bu* & *ma* and question

¹⁵ Without the aid of phonetic experiments, it is impossible to determine whether rising declarative intonation co-occurs with other question markers. We therefore leave its (non-)co-occurrence pattern for future work.

Table 4 Co-occurrence of question markers

	<i>Wh-</i>	<i>Meiyou</i>	A-not-A	<i>Bu</i>	<i>Ma</i>	<i>Ne</i>	<i>Ba</i>
<i>Ba</i>	✗	✗	✗	✗	✗	✗	
<i>Ne</i>	✓	✓	✓	✗	✗		
<i>Ma</i>	✗	✗	✗	✗			
<i>Bu</i>	✗	✗	✗				
A-not-A	✗	✗					
<i>Meiyou</i>	✗						
<i>Wh-</i>							

markers in ForceP, as well as between *ba* and all other question markers, it calls for a new analysis of their complementary distribution. We attribute their non-co-occurrence to their semantics: the output of a lower question marker is not of the right type to serve as an input of any higher question markers. Deferring a deeper discussion to later sections, we briefly note here that we adopt a [Karttunen \(1977\)](#) analysis for question semantics: we assume that a proposition is of type st , while a question is of type $\langle st, t \rangle$ (i.e., a set of propositions). Moreover, we propose that the *wh*-operator, A-not-A, *meiyou*, *bu*, and *ma* take propositions and yield questions, while *ne* takes and yields questions. As for *ba*, we propose that it takes and yields propositions based on its assertoric nature argued by [X. Yuan \(2021\)](#) and [Ye \(2024\)](#), a point we elaborate on in [Section 5.2](#). We summarize the semantic types of the input and output of question markers in [Table 5](#).

Table 5 Semantic types of the input and output of question markers

ForceP			PerspP		CommitP
$st \rightarrow$	$wh-$	$\rightarrow \langle st, t \rangle$	$st \rightarrow$	$bu \rightarrow \langle st, t \rangle$	$st \rightarrow ba \rightarrow st$
$st \rightarrow$	$meiyou$	$\rightarrow \langle st, t \rangle$	$st \rightarrow$	$ma \rightarrow \langle st, t \rangle$	
$st \rightarrow$	A-not-A	$\rightarrow \langle st, t \rangle$	$\langle st, t \rangle \rightarrow$	$ne \rightarrow \langle st, t \rangle$	

Since all question markers except for *ne* take propositions, while all question markers except for *ba* yield sets of propositions, they cannot co-occur with each other due to type clash. With respect to *ba*, although it yields propositions, which are type-wise acceptable inputs to other question markers, it belongs to the highest one among the three projections and cannot feed its output to a lower question marker. The upshot is that the non-co-occurrence of question markers can be accounted for under our proposal as long as we assume a distinction in semantic representation between propositions and questions.

4.2 Semantic component: P-activeness

The second component of our analysis argues that PerspP should be interpreted à la [Dayal 2025](#) in the sense that it triggers perspectival centering and introduces a requirement on the embedded question’s potential activeness for the perspectival center. In [Dayal’s \(2025\)](#) proposal, the perspective head in interrogatives poses a not-at-issue requirement that the complement question must be potentially active (p-active) for the perspectival center. Two concepts merit further clarification: *perspectival center* and *p-activeness*.

Roughly speaking, the perspectival center is “the one who holds the question.” In Dayal’s implementation, the perspectival center is syntactically introduced in the specifier position of PerspP by a phonologically null anaphor (i.e., PRO) that must be bound. In a matrix question, this anaphor is bound by the speaker. When the PerspP is embedded as in (45), the anaphor is bound by a matrix argument, typically the matrix subject.

(45) *Embedded PerspP: Perspectival center = matrix argument*

(The question of whether it rained is p-active for the matrix subject John.)

John_i xiang zhidao [_{PerspP} PRO_i [_{Persp'} xia yu le [_{Persp⁰} ma]]]_↑.
John want know fall rain PRF MA

‘John wants to know whether it rained.’

(Bhatt & Dayal 2020: 1122, adapted)

For a question Q to be p-active for the perspectival center x , a minimal requirement is the possibility of x ’s ignorance about the answer to Q as in (46), or the IR as discussed in Section 3.1.¹⁶ This not-at-issue requirement is formalized as a lexical property of Persp⁰_{CQ}, as copied in (47), where CQ stands for *centered question*.¹⁷

(46) *Minimal requirement for p-activeness*

$\forall Q [\forall x [\text{p-active-for}(Q, x) \rightarrow \Diamond \neg \text{know}(x, \text{answer}(Q))]]$ (Dayal 2025: 673)

(47) *Denotation of Persp⁰_{CQ}*

$\llbracket \text{Persp}_{\text{CQ}}^0 \rrbracket = [\lambda Q. \lambda x: \text{p-active-for}(Q, x). Q]$ (Dayal 2025: 673)

A key point of divergence between our framework and Dayal’s lies in the input type of Persp⁰_{CQ}. In Dayal’s system, Persp⁰_{CQ} in interrogatives takes questions as its input, as interrogative clause-typing uniformly takes place at CP (roughly corresponding to our ForceP). In contrast, for Mandarin PerspP question markers, we slightly modify the denotation to allow certain markers to take propositions as input, and we justify this move in Section 6. This is illustrated in (48) using *ma* as an example, where we use PR(x) to abbreviate the p-activeness requirement for x .¹⁸

(48) $\llbracket \text{ma} \rrbracket = [\lambda p. \lambda x: \text{PR}(x). [\lambda q. q = p]]$

The paradigmatic distinctions between QS and NQS (or between IR satisfied and IR not satisfied) in their embedding of *bu* and *ma* follow from the PR. Since *bu* and *ma* realize Persp⁰, they introduce the PR when embedded, with the matrix subject being the perspectival center. In a matrix declarative (without modality or negation), the PR cannot be satisfied by an NQS due to its semantics, whereas it can be met by a QS:

¹⁶ Apart from the IR as a minimal requirement, possible candidates for the PR’s pragmatic component include *resolution requirement* (i.e., “whether a positive answer to the matrix question in each case can lead to a resolution of the question that the speaker is interested in” (Dayal 2025: 685)), *knowledge requirement* (i.e., whether “[i]t is possible that the addressee knows the answer” (Y. Liu 2025: 93)), or the perspectival center’s curiosity toward the question, which may help account for the unacceptability of examples such as (20b).

¹⁷ Dayal (2025) suggests that the three-layered peripheral structure is present in declaratives and imperatives as well, where the perspective head is realized as Persp⁰_{CP} (*centered proposition*) and Persp⁰_{CI} (*centered imperative*), respectively. We adopt these variants of the perspective head at face value and refer interested readers to Dayal 2025.

¹⁸ In (48), we follow Dayal’s (2025) treatment of p-activeness as a presupposition, but we acknowledge that the not-at-issue requirement may behave more like a post-supposition (e.g., Brasoveanu 2013, Charlow to appear), since it is possibly not evaluated at the point of composition.

(49) #John zhidao [xia yu le **ma**]_(↑). NQS: ✗PR

John know fall rain PRF MA

Intended: ‘John knows whether it rained.’

- a. John knows the answer to the question of whether it rained.
- b. It is not possible that John does not know the answer.
- c. The question of whether it rained is not p-active for John.

(50) John xiang zhidao [xia yu le **ma**]_(↑). QS: ✓PR

John want know fall rain PRF MA

‘John wants to know whether it rained.’

(Bhatt & Dayal 2020: 1122, adapted)

- a. John wants to know the answer to the question of whether it rained.
- b. It is possible that John does not know the answer.
- c. The question of whether it rained is p-active for John.

Concretely, we propose that question embedding predicates, be they QS or NQS, can freely select a ForceP or a PerspP. There is nothing in the syntax that prevents a PerspP from being the complement of either type. Yet, an NQS is incompatible with contexts in which the centered question is p-active for the perspectival center: one cannot possibly be ignorant about the answer to a question when they already know it.

4.3 Necessity for both components

The presence of the PR demands a conspiratorial theory of selection, where syntax and semantics both contribute to the (un)acceptability of clausal complementation. Besides the effect of matrix modality, negation, and interrogative discussed in Section 3.1, another piece of evidence that semantics plays a role in interrogative complementation has to do with the fact that the way to say ‘wonder’ in Mandarin is lexically composed of ‘want’ and ‘know’. Should selection be purely syntactic here, ‘want to know’ would share the same embedding profile as ‘know’. This has shown to be not true, as we have seen in Section 3.1 and Section 4.2, where QS and NQS differ in their sensitivity to the PR.

The semantic component of our proposal does not undermine the necessity for a syntactic distinction between ForceP and PerspP. An alternative account that avoids the splitting of PerspP from ForceP would predict that *bu* and *ma* questions can be embedded in any question embedding context as long as the context satisfies the PR, including noun complements and clausal subjects. This prediction is not borne out. In (51), for instance, the perspectival center — in this case, the speaker — is ignorant of the answer to the question, but the sentence is still unacceptable.

(51) *Noun complement or clausal subject*

*[Mingtian xia yu **bu/ma**]_(↑) (de wenti) hai bu qingchu.

tomorrow fall rain BU MA COMP question yet not clear

Intended: ‘(The question of) whether it will rain tomorrow is not clear yet.’

The same observation can be made with clausal objects of prepositions and with unconditionals. The p-activeness of the embedded question for the question holders in (52–53) does not rescue the sentence from the unacceptability of embedding *ma* questions.

(52) *Clausal object of a preposition*

Xiaohong dui [zuotian xia-le yu **meiyou/*bu/*ma**] hen haoqi.
 Xiaohong about tomorrow fall-PFV rain MEIYOU BU MA very curious
 (Intended:) ‘Xiaohong is very curious about whether it will rain tomorrow.’

(53) *Unconditional*

Wulun [wo jige-le **meiyou/*bu/*ma**], wo dou xiwang ni gaosu wo.
 no.matter 1SG pass-PFV MEIYOU BU MA 1SG all hope 2SG tell 1SG
 (Intended:) ‘No matter if I passed, I hope you will tell me.’

In the current account, we attribute the observations in (51–53) to the requirement on the clause size in the corresponding question embedding contexts: a PerspP is too large for these contexts to embed. Consequently, *bu* and *ma* are banned from these contexts, regardless of the epistemic status of the perspectival center.

4.4 Derivations of the question markers’ (un)embeddability

With the two components established, we derive the (un)embeddability patterns of all aforementioned question markers. *Wh*-questions, *meiyou*, and A-not-A are all completely embeddable. They do not introduce the PR, and we consider them to be of the ForceP size. The derivation of their embeddability is illustrated using *meiyou* in (54–55).

(54) *Embedded meiyou question as noun complement or clausal subject*

[Zuotian xia-le yu **meiyou**] (de wenti) hen guanjian.
 Zhangsan fall-PFV rain COMP question very crucial
 ‘(The question of) whether it rained yesterday is very crucial.’

- a. Noun complements and clausal subjects must be no larger than ForceP.
- b. A *meiyou* question is a ForceP and not larger than ForceP (✓syntax).
- c. (54) is acceptable.

(55) *Embedded meiyou question as clausal object*

Xiaohong_i (xiang) zhidao [ta_i jige **meiyou**].
 Xiaohong want know 3SG pass MEIYOU
 ‘Xiaohong_i knows/wants to know whether she_i passed.’

- a. Clausal objects of question embedding predicates must be no larger than PerspP.
- b. A *meiyou* question is a ForceP and not larger than PerspP (✓syntax).
- c. (55) is acceptable.

As for *bu*, *ma*, and *ne* questions, they are not embeddable as noun complements or clausal subjects due to their size being PerspPs and exceeding ForcePs. When embedded as clausal objects, they introduce the PR. We illustrate their derivation using *ma* in (56–58).

(56) *Embedded ma question as noun complement or clausal subject*

*[Zuotian xia-le yu **ma**]_(T) (de wenti) hen guanjian.
 yesterday fall-PFV rain MA COMP question very crucial
 Intended: ‘(The question of) whether it rained yesterday is very crucial.’

- a. Noun complements and clausal subjects must be no larger than ForceP.
 - b. A *ma* question is a PerspP and larger than ForceP (~~X~~_{syntax}).
 - c. (56) is unacceptable.
- (57) *Embedded ma question as clausal object of NQS (in a matrix declarative without modality or negation)*
 #Xiaohong_i zhidao [ta_i de fenshu bi wo_j gao **ma**]_(↑).
 Xiaohong know 3SG POSS score than 1SG high MA
 Intended: ‘Xiaohong_i knows whether her_i score is higher than mine_j.’
- a. Clausal objects of question embedding predicates must be no larger than PerspP.
 - b. A *ma* question is a PerspP and not larger than PerspP (✓_{syntax}).
 - c. A *ma* question introduces the PR.
 - d. Since the matrix subject knows the answer, the PR is not satisfied (~~X~~_{semantics}).
 - e. (57) is unacceptable.
- (58) *Embedded ma question as clausal object of QS (in an matrix declarative without modality or negation)*
 Xiaohong_i xiang zhidao [ta_i de fenshu bi wo_j gao **ma**]_(↑).
 Xiaohong want know 3SG POSS score than 1SG high MA
 ‘Xiaohong_i wants to know whether her_i score is higher than mine_j.’
- a. Clausal objects of question embedding predicates must be no larger than PerspP.
 - b. A *ma* question is a PerspP and not larger than PerspP (✓_{syntax}).
 - c. A *ma* question introduces the PR.
 - d. Since the matrix subject wants to know the answer, the PR is satisfied (✓_{semantics}).
 - e. (58) is acceptable.
- Ba* questions are unembeddable across the board. We attribute their unembeddability to their exceedingly large clause size and illustrate this in (59–60).
- (59) *Embedded ba question as noun complement or clausal subject: X*
 *[Zuotian xia-le yu **ba**]_(↑) (de wenti) hen guanjian.
 yesterday fall-PFV rain BA COMP question very crucial
 Intended: ‘(The question of) whether it rained yesterday is very crucial.’
- a. Noun complements and clausal subjects must be no larger than ForceP.
 - b. A *ba* question is a CommitP and larger than ForceP (~~X~~_{syntax}).
 - c. (59) is unacceptable.
- (60) *Embedded ba question as clausal object: X*
 *Xiaohong_i (xiang) zhidao [ta_i de fenshu bi wo_j gao **ba**]_(↑).
 Xiaohong want know 3SG POSS score than 1SG high BA
 Intended: ‘Xiaohong_i knows/wants to know whether her_i score is higher than mine_j.’
- a. Clausal objects of question embedding predicates must be no larger than PerspP.
 - b. A *ba* question is a CommitP and larger than PerspP (~~X~~_{syntax}).
 - c. (60) is unacceptable.

5 Different wines in different vessels

We supplement our split iForceP proposal and strengthen our semantic account of the non-co-occurrence of question markers by fully articulating our implementation of their semantics. Ultimately, we arrive at the paradigm presented in Table 6. Along the way, we present evidence that any plausible implementation of the semantics of Mandarin question markers should reflect two distinctions — namely, (i) an assertion-question distinction (as discussed in Section 4.1) and (ii) a monopolar-bipolar distinction in polar questions. The first two upcoming subsections serve the first point, and the next two serve the second.

Table 6 Inputs and outputs of question markers
Note: Q_{PL} represents a plural propositional set.

ForceP	PerspP	CommitP
$p \rightarrow wh- \rightarrow Q_{\text{PL}}$	$p \rightarrow bu \rightarrow \{p, \neg p\}$	$p \rightarrow ba \rightarrow p$
$p \rightarrow meiyou \rightarrow \{p, \neg p\}$	$p \rightarrow ma \rightarrow \{p\}$	
$p \rightarrow A\text{-not-}A \rightarrow \{p, \neg p\}$	$Q_{\text{PL}} \rightarrow ne \rightarrow Q_{\text{PL}}$	

We begin by introducing five documented assertiveness diagnostics and extending them to Mandarin, arguing that *ba* is assertoric in nature in comparison with rising declaratives and other canonical question-forming strategies. We then use Bolinger contexts and other pieces of language-internal empirical evidence to show how the distributional distinction between *ma* questions and *meiyou/A-not-A/bu* questions resemble the monopolar-bipolar distinction crosslinguistically attested.

5.1 Assertiveness and nonassertiveness

We should now ask ourselves a philosophical question: What is an assertion? In response to this question, philosophers of language have provided us with a list of interrelated properties for canonical assertions, which are used to design assertiveness diagnostics.

First, a canonical assertion commits the speaker to its propositional content (e.g., Searle 1979, Krifka 2015). Thus, when a speaker makes two assertions that have opposite propositional contents, a contradiction arises, and the discourse is considered aberrant (61a). On the other hand, canonical questions — genuine information-seeking questions that carry no bias — do not make public what the speaker commits. In this case, even if two canonical polar questions have completely incompatible sentence radicals, they can be uttered in a sequence (61b).

(61) *Commitment*

- a. John is in Beijing. #John is in Shanghai.
- b. Is John in Beijing? Is John in Shanghai?

canonical assertion: ✓

canonical question: ✗

Second, canonical assertions are capable of, and are typically used for, proffering information (Bach & Harnish 1979). This specific type of information proffering, which Roberts (2012, 2023) refers to as *proposition proffering*, allows certain discourse moves by the addressee that are not allowed by questions. For example, Rudin (2022) observes that English assertions but not polar questions can be replied with an expression of grat-

itude for the speaker's act of giving the information (e.g., “thanks for the heads up”) or with an indication of receiving the previously unknown information (e.g., “oh, I had no idea”), as shown in (62–64).¹⁹

(62) *Context for (63–64)*

(Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she's been fired from her job. He turns to Bertha, who is close with Carrie, and says: ...)

(Rudin 2022: 344)

(63) *Canonical assertion: ✓ Proposition proffering*

A: Carrie got fired.

a. B: Thanks for the heads up.

b. B: Oh, I had no idea.

(Rudin 2022: 344–345)

(64) *Canonical question: ✗ Proposition proffering*

A: Did Carrie get fired?

a. B: #Thanks for the heads up.

b. B: #Oh, I had no idea.

(Rudin 2022: 344)

Third, a canonical assertion assumes that the speaker obtains knowledge about or at least belief in what they assert (Williamson 1996). As such, when an assertion is accepted, the addressee is not obliged to provide more information. A canonical question, on the other hand, does request for more information, and once accepted, it puts the addressee under the obligation to provide as much information as they can to resolve it (Roberts 2012). Rudin (2022) tests this contrast by setting up a context in which the addressee raises a related issue in response to the speaker's utterance (65). In this configuration, the addressee's acceptance of the move made by the speaker is indicated by the absence of overt rejection. When the speaker utters an assertion, raising a related issue is considered natural (66), whereas when the speaker utters a question, the same response is considered unacceptable (67). Rudin therefore concludes that questions but not assertions solicit answers.

(65) *Context for (66–67)*

(Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she has a new girlfriend. He turns to Bertha, who is close with Carrie, and says: ...)

(Rudin 2022: 345)

¹⁹ There are cases where a question seems to be used for proffering information, too. A classic example is below:

(i) By the way, did you know that Jack Robinson was my cousin?

(Bolinger 1979: 88)

Bolinger's (1979) example can apparently be responded with “oh, I had no idea,” but it cannot be responded with “thanks for the heads up.” Anticipating the diagnostics introduced in the upcoming Mandarin subsection, we also note that “I see” is not a felicitous response. In light of these observations, the felicity of “oh, I had no idea” may arise either from accommodation or from the recognition that it resolves the matrix question to some extent, since it entails that the responder does not know the answer. Bolinger also observes that this type of noncanonical questions typically occur with factives. In this section, we avoid using factives in our examples so as not to trigger such noncanonical interpretations.

(66) *Canonical assertion: ✗Answer solicitation*

A: Carrie got a new girlfriend.

- a. B: Yeah, she told me about it this morning.
- b. B: I don't think so. Maybe she's just trying to stir up drama.
- c. B: Did you know Delia is leaving her husband?

(Rudin 2022: 345)

(67) *Canonical question: ✓Answer solicitation*

A: Did Carrie get a new girlfriend?

- a. B: Yeah, she told me about it this morning.
- b. B: I don't think so. Maybe she's just trying to stir up drama.
- c. B: #Did you know Delia is leaving her husband?

(Rudin 2022: 345)

For the sake of completeness, we draw on two more diagnostics frequently applied to differentiate assertions and questions: the 'after all' test and the 'tell me' test (Sadock 1971, 1974, Asher & Reese 2005, 2007, Reese & Asher 2007, 2009, Kamali 2020, X. Yuan 2021). Based on the observation by Sadock (1971: 225–227) that *after all* co-occurs with declaratives much more freely than questions, Asher & Reese (2005, 2007) and Reese & Asher (2007, 2009) generalize that sentence-initial *after all* can precede assertions (68a) but not neutral questions (68b).

(68) *'After all'*

It is fine if you don't finish the article today.

- a. After all, your adviser is out of the country.
- b. #After all, is your adviser out of the country?

*canonical assertion: ✓**canonical question: ✗*

(Reese & Asher 2009: 157)

As for the 'tell me' test, Sadock (1974: 112) observes that true questions allow this imperative pretag (69a), whereas assertions do not (69b).

(69) *'Tell me'*

- a. #Tell me, John owns a car.
- b. Tell me, does John own a car?

*canonical assertion: ✗**canonical question: ✓*

(Reese & Asher 2009: 158)

Each of these diagnostics should be applied with caution, especially when not accompanied by other tests, but when considered together, their results help reveal the semantic flavor of different structures. A summary of the diagnostics covered in this subsection is presented in Table 7, where we add results of applying these diagnostics to English tag questions (cf. Asher & Reese 2007, Reese & Asher 2007, 2009) and RDs, which will be relevant to the discussion in Section 5.2.

Table 7 Assertiveness diagnostics in English

	Commitment	Proposition proffering	Answer solicitation	‘After all’	‘Tell me’
Canonical assertion	✓	✓	✗	✓	✗
Canonical question	✗	✗	✓	✗	✓
Tag question	✗	✗	✓	✓	✓
RD	✗	✗	✓	✗	✗

5.2 Applying assertiveness diagnostics in Mandarin

In this subsection, we apply the assertiveness diagnostics in Mandarin. As the semantic nature of different Mandarin question-forming strategies is now at issue, while the statuses of declaratives, *wh*-questions, and nonconfirmation polar questions are relatively clear in the literature, *ba* and RDs require further investigation. For this reason, we refrain from providing free translations for examples of *ba* and RDs in this subsection, even though we have done so in previous sections to illustrate their uses. Additionally, anticipating our conclusion, we make the following adjustments and clarifications to our presentation and terminology related to *ba*: (i) we replace the term “*ba* questions” with “*ba* utterances”; and (ii) we punctuate *ba* utterances with full stops, but we do not mean to suggest any distinction in the use of *ba* between the *ba* utterance examples in this subsection and the *ba* confirmation question examples discussed thus far.

We begin with the commitment test, where we provide a declarative (70a) and a *ma* question (70b) as baselines. In terms of (non)contradiction, we observe that *ba* utterances (70c) behave like declaratives, whereas RDs (70d) behave like *ma* questions. (See also Yang 2021: 529 and X. Yuan 2021: 912 for other examples of this test applied to *ba*.)

(70) Commitment

- a. Xiaoming zai Beijing. #Xiaoming zai Shanghai. declarative: ✓
Xiaoming at Beijing Xiaoming at Shanghai
‘Xiaoming is in Beijing. #Xiaoming is in Shanghai.’
- b. Xiaoming zai Beijing **ma**? Xiaoming zai Shanghai **ma**? ma question: ✗
Xiaoming at Beijing MA Xiaoming at Shanghai MA
‘Is Xiaoming in Beijing? Is Xiaoming in Shanghai?’
- c. Xiaoming zai Beijing **ba**. #Xiaoming zai Shanghai **ba**. ba utterance: ✓
Xiaoming at Beijing BA Xiaoming at Shanghai BA
- d. Xiaoming zai Beijing? Xiaoming zai Shanghai? RD: ✗
Xiaoming at Beijing Xiaoming at Shanghai

As for proposition proffering, the paradigm we observe in (70) is fully replicated as in (71–75). Below, we use ‘thanks for letting me know’ as an expression of gratitude toward the speaker, and we use ‘I see’ as an indication of receiving previously unknown information.

(71) *Context for (72–75)*

(Angela is looking at WeChat on her phone, where she sees a cryptic post by her friend Xiaoming, which seems to suggest the he got married. She turns to Barbie, who is close with Xiaoming, and says: ...)

(72) *Declarative: ✓Proposition proffering*

A: Xiaoming jiehun le.

Xiaoming marry PRF

‘Xiaoming got married.’

a. B: *Duoxie gaozhi*. ‘Thanks for letting me know.’

b. B: *Yuanlai ruci*. ‘I see.’

(73) *Ma question: ✗Proposition proffering*

A: Xiaoming jiehun le **ma**?

Xiaoming marry PRF MA

‘Did Xiaoming get married?’

a. B: *#Duoxie gaozhi*. ‘Thanks for letting me know.’

b. B: *#Yuanlai ruci*. ‘I see.’

(74) *Ba utterance: ✓Proposition proffering*

A: Xiaoming jiehun le **ba**.

Xiaoming marry PRF BA

a. B: (?)*Duoxie gaozhi*. ‘Thanks for letting me know.’

b. B: *Yuanlai ruci*. ‘I see.’

(75) *RD: ✗Proposition proffering*

A: Xiaoming jiehun le?

Xiaoming marry PRF

a. B: *#Duoxie gaozhi*. ‘Thanks for letting me know.’

b. B: *#Yuanlai ruci*. ‘I see.’

The same distinction exists in answer solicitation, but the direction of the pattern is reversed. In the same context as in (65), copied below in (76), declaratives (77) and *ba* utterances (79) do not solicit answers, but *ma* questions (78) and RDs (80) do.

(76) *Context for (77–80)*

(Alvin is looking at Facebook on his phone, where he sees a cryptic post by his friend Carrie, which seems to suggest that she has a new girlfriend. He turns to Bertha, who is close with Carrie, and says: ...)

(Rudin 2022: 345)

(77) *Declarative: ✗Answer solicitation*

A: Carrie jiao-le ge xin nü pengyou.

Carrie get-PFV CLF new girl friend

‘Carrie got a new girlfriend.’

a. B: *Queshi*. ‘Indeed.’

b. B: *Wo tingshuo Delia ye jiao-le ge xin nü pengyou*. ‘I heard that Delia also got a new girlfriend.’

(78) *Ma question: ✓ Answer solicitation*A: Carrie jiao-le ge xin nü pengyou **ma**?

Carrie get-PFV CLF new girl friend MA

'Did Carrie get a new girlfriend?'

a. B: *Queshi*. 'Indeed.'b. B: #*Wo tingshuo Delia ye jiao-le ge xin nü pengyou*. 'I heard that Delia also got a new girlfriend.'(79) *Ba utterance: ✗ Answer solicitation*A: Carrie jiao-le ge xin nü pengyou **ba**.

Carrie get-PFV CLF new girl friend BA

a. B: *Queshi*. 'Indeed.'b. B: *Wo tingshuo Delia ye jiao-le ge xin nü pengyou*. 'I heard that Delia also got a new girlfriend.'(80) *RD: ✓ Answer solicitation*

A: Carrie jiao-le ge xin nü pengyou?

Carrie get-PFV CLF new girl friend

a. B: *Queshi*. 'Indeed.'b. B: #*Wo tingshuo Delia ye jiao-le ge xin nü pengyou*. 'I heard that Delia also got a new girlfriend.'

Finally, the 'after all' (81) and 'tell me' (82) tests have been applied to Mandarin polar questions and *ba* utterances by X. Yuan (2021). We add examples of declaratives and RDs.

(81) 'After all'

a. Bijing, ni mingtian yao qu xuexiao.

after.all 2SG tomorrow will go school

'After all, you will go to school tomorrow.'

*declarative: ✓*b. #Bijing, ni mingtian yao qu xuexiao **ma**?

after.all 2SG tomorrow will go school MA

'After all, will you go to school tomorrow?'

ma question: ✗

(X. Yuan 2021: 911, adapted)

c. Bijing, ni mingtian yao qu xuexiao **ba**.

after.all 2SG tomorrow will go school BA

ba utterance: ✓

(X. Yuan 2021: 911)

d. #Bijing, ni mingtian yao qu xuexiao?

after.all 2SG tomorrow will go school

RD: ✗

(82) 'Tell me'

a. #Gaosu wo, ni mingtian yao qu xuexiao.

tell 1SG 2SG tomorrow will go school

'Tell me, you will go to school tomorrow.'

*declarative: ✗*b. Gaosu wo, ni mingtian yao qu xuexiao **ma**?

tell 1SG 2SG tomorrow will go school MA

'Tell me, will you go to school tomorrow?'

ma question: ✓

(X. Yuan 2021: 911, adapted)

c. #Gaosu wo, ni mingtian yao qu xuexiao **ba**.

tell 1SG 2SG tomorrow will go school BA

ba utterance: ✗

(X. Yuan 2021: 911)

d. #Gaosu wo, ni yao qu xuexiao? RD: ✗
 tell 1sg 2sg tomorrow will go

We summarize (70–82) in Table 8, where we add results for *meiyou*, A-not-A, *bu*, and *ne* questions.²⁰ We draw three conclusions from these observations. First, *ba* utterances share their distributional pattern with canonical assertions, supporting a unified analysis of *ba*₂ (discussed in Section 2) and the confirmation-seeking use of *ba* (e.g., X. Yuan 2021), and they do not follow the pattern of canonical questions or English tag questions at all, the latter of which are frequently used to translate confirmation-seeking *ba* utterances. Second, Mandarin RDs resemble canonical questions in both languages and share the same properties as English RDs. Last but not least, *ba* utterances should not be interpreted as conveying a pragmatic flavor similar to that of RDs, supporting Yang (2021) and X. Yuan (2021) but contrary to some other previous claims (e.g., Han 1995). Together, these observations support our implementation in which questions formed with *meiyou*, A-not-A, *bu*, *ma*, *ne*, or RDs are analyzed as sets of propositions, while declaratives and *ba* utterances are analyzed as propositions. From this point and onward, we refer to *ba* utterances as “*ba* declaratives.”

Table 8 Assertiveness diagnostics in Mandarin

	Commitment	Proposition proffering	Answer solicitation	‘After all’	‘Tell me’
Declarative	✓	✓	✗	✓	✗
<i>Meiyou</i> question	✗	✗	✓	✗	✓
A-not-A question	✗	✗	✓	✗	✓
<i>Bu</i> question	✗	✗	✓	✗	✓
<i>Ma</i> question	✗	✗	✓	✗	✓
<i>Ne</i> question	✗	✗	✓	✗	✓
<i>Ba</i> utterance	✓	✓	✗	✓	✗
RD	✗	✗	✓	✗	✗

5.3 Singleton and doubleton

We move on to establish the monopolar-bipolar distinction in Mandarin polar questions. In other languages that have been argued to exhibit this distinction, the empirical support comes from observed contrasts in (i) the (in)compatibility with unconditionals (e.g., Biezma & Rawlins 2012, Dayal 2025) and (ii) the (un)acceptability in a specific type of contexts (e.g., Krifka 2015, Matthewson 2025, Kamali 2025). The first group of empirical support is not applicable in Mandarin, given that a subset of Mandarin question markers cannot embed under unconditionals due to independent syntactic constraints, as discussed in Section 4.3. This subsection therefore mainly focuses on the second group of empirical support.

²⁰ Due to restrictions to be discussed in Section 5.3, *meiyou*, A-not-A, *bu*, and *ne* questions cannot be felicitously uttered in the contexts we provide for proposition proffering and answer solicitation in this subsection. In Table 8, the judgments for these question markers in these two tests are based on a controlled context in which both these questions and *ma* questions can be felicitously uttered—for example, a context where the speaker has no direct or indirect evidence and is genuinely wondering about the question they ask. A controlled context in which both questions with these question markers and declaratives/*ba* utterances/RDs can be felicitously uttered is impossible to construct.

Observations about the distributional distinction between monopolar and bipolar questions date back to Bolinger (1979), who lists contexts where a plain English polar question behaves differently from its counterpart in the form of an alternative question.²¹ Contexts that exhibit this distinction are referred to as *Bolinger contexts*, and we follow Matthewson (2025) and abbreviate the term as *B-contexts*. An example B-context is presented in (83), where the context provides evidence against a negative response. Another example B-context is (84), where a positive response is the most likely, but the other alternatives are possible as well.

- (83) (Somebody in your house has announced they're leaving. A little while later, you run into them in the kitchen, and you are very surprised, and you say: ...) (Matthewson 2025: 30)
Are you still around (*or not)? (Bolinger 1979: 88)
- (84) (Your colleague is grumpy today, and they look not great. You ask them: ...) (Matthewson 2025: 33)
What's the matter? Are you tired (*or not)? (Bolinger 1979: 89)

To capture the contrast between polar questions and their alternative question counterparts, a line of work (van Rooij & Šafářová 2003, AnderBois 2011, Biezma & Rawlins 2012, Krifka 2015, 2022, 2024, Tabatowski 2022, Kamali 2025, Matthewson 2025) proposes representing the former as singletons (i.e., monopolar $\{p\}$) and the latter as doubletons (i.e., bipolar $\{p, \neg p\}$). The core insight is that a question does not have to denote all of its possible answers (as suggested in Hamblin 1973, for instance); rather, it denotes a set of alternative(s) that the speaker hopes to make salient (Biezma & Rawlins 2012). By asking a question, the speaker invites the answerer to choose between the salient alternatives provided by both the context and the denotation of the question. To illustrate, we begin by adopting Biezma & Rawlins's (2012) question operator, defined in (85–86).

- (85) *Question operator*
 $\llbracket [_{\text{Q}}] \alpha \rrbracket^c = \llbracket \alpha \rrbracket^c$, defined only if
 a. $\llbracket \alpha \rrbracket^c \subseteq \text{SalientAlts}(c)$ or $\text{SalientAlts}(c) = \emptyset$, and
 b. $|\llbracket \alpha \rrbracket^c \cup \text{SalientAlts}(c)| > 1$.
 (Biezma & Rawlins 2012: 392)
- (86) *Definition of SalientAlts*
 $\text{SalientAlts}(c)$ is the set of propositional alternatives that are salient in the context of interpretation c .
 (Biezma & Rawlins 2012: 388)

As (85b) suggests, a question can only be felicitously asked if the context and the denotation of the question together offer more than one alternative. In the case of a bipolar question, this part of the presupposition is trivially satisfied by the two members within the alternative set represented by the question itself, but in a monopolar question, the felicity relies on the salient alternatives implicitly inferred from the context. As Matthewson (2025) puts it, a monopolar question has the alternative structure $\{p, q, r, \dots\}$, whereas a bipolar

²¹ In English, an *or not* question triggers a *cornering effect* (Biezma 2009), urging the addressee to give a response, an effect not found with plain polar questions. To avoid this confound in B-contexts, in place of an *or not* question, one could instead use a *complement alternative question* (CAQ), formed by two alternatives that take up the whole possibility space (e.g., “Is the light on or off?”, “Are you tired or still doing alright?”, etc.). CAQs do not trigger the cornering effect (Beltrama, Meertens & Romero 2020) and still show a contrast with plain polar questions. This confound can also be avoided in languages where the cornering effect is encoded by a separate marker, and one such example is Thompson River Salish, where the cornering marker is argued to be bipolar, too (Matthewson 2025). As for Mandarin, while *ne* following a polar question has been reported to resemble English *or not* questions, as discussed in Section 3.3, we observe no other question markers examined in this paper to give rise to a cornering effect. We also discuss another SFP that has been described as producing an effect similar to cornering in Section 5.4.

question has the alternative structure $\{p, \neg p\}$, whose members take up the whole possibility space. This is illustrated in (87).

(87) (It has been established that there are three epistemically possible meals that the addressee may cook: pasta, stew, and fish.)

a. *Monopolar question*
Are you making pasta?

b. *Bipolar question*
Are you making pasta or not?

SalientAlts(c)

you are making pasta
you are making stew (implicit)
you are making fish (implicit)

SalientAlts(c)

you are making pasta
you are not making pasta

(Biezma & Rawlins 2012: 399–400)

Within this framework, (85a) predicts that a bipolar question cannot be felicitously asked in a context where exactly one of the two alternatives represented in its denotation is salient in the context. The two B-contexts in (83–84) are such contexts: in each, p is salient, but $\neg p$ is not. This accounts for why *or not* questions are not felicitous in these B-contexts, while plain polar questions are.

5.4 B-contexts and more in Mandarin

Similar contrasts under B-contexts are attested in Thompson River Salish (Matthewson 2025) and Turkish (Kamali 2025), as well as in Mandarin. Krifka (2015) reports a contrast between *ma* and A-not-A questions in a B-context where a $\neg p$ response is blocked (88), a context that resembles (83).

(88) (The speaker sees the addressee eating an apple.)

a. *Ma question: ✓*
Ni chi pingguo **ma**?
2SG eat apple MA
'Do you eat apples?'

b. *A-not-A question: ✗*
#Ni **chi-bu-chi** pingguo?
you eat-not-eat apple
'Do you eat apples?'

(Krifka 2015: 11)

Krifka's observation is corroborated by Yuan & Hara's (2019) survey of sixteen native Mandarin speakers. They report that the participants have no preference between *ma* and A-not-A questions when the context accepts both p and $\neg p$ as possible answers, but they have strong preference for *ma* questions when the context blocks $\neg p$ from being a possible answer.

Prompted by Krifka and Yuan & Hara, we replicate in Mandarin the B-context contrasts observed in (83–84) below in (89–90).

(89) (Somebody in your house has announced they're leaving. A little while later, you run into them in the kitchen, and you are very surprised, and you say: ...)

(Matthewson 2025: 30)

a. *Ma question: ✓*
Ni hai zai zhe **ma**?
2SG still at here MA
'Are you still here?'

b. *A-not-A question: ✗*
#Ni hai **zai-bu-zai** zhe?
2SG still at-not-at here
'Are you still here?'

(90) (Your colleague looks pale today. You ask them: ...)

- | | |
|--|---|
| <p>a. <i>Ma question:</i> ✓
 Ni bing-le ma?
 2SG get.sick-PFV MA
 ‘Did you get sick?’</p> | <p>b. <i>A-not-A question:</i> ✗
 #Ni bing-mei-bing?
 2SG get.sick-not.PFV-get.sick
 ‘Did you get sick?’</p> |
|--|---|

With respect to the (un)acceptability under B-contexts, while RDs follow *ma* questions, *meiyou*, *bu*, and *ne* questions all pattern identically to A-not-A questions. This dichotomy lends support to the analysis that *ma* questions and RDs are monopolar, while questions formed with the other question markers are bipolar.

Aside from contrasts observed in B-contexts, we present three distributional facts in Mandarin that a monopolar-bipolar distinction naturally accounts for. These distributional facts pertain to (i) the SFP *a*, (ii) the bias marker *nandao*, and (iii) ‘right’ as a response.²²

The SFP *a* (which may be realized as *na*, *nga*, *ra*, *wa*, *ya*, or *za* in different phonological contexts) conveys the speaker’s attitude and emotion (Zhu 1982). Among other usages, *a* can occur in a question and convey that the speaker is requiring an answer (Chu 2009: 284–285), an effect somewhat similar to the cornering effect in English *or not* questions (Biezma 2009). As presented in (91), *a* cannot occur with *ma* or *ba*, but it can occur with *wh*-, *meiyou*, A-not-A, *bu*, and *ne* questions.

- (91) a. Baozhi **shenme shihou** lai a?
 newspaper what time come A
 ‘When will the newspaper arrive?’
- b. Baozhi lai-le **meiyou/bu/*ma/*ba** a?
 newspaper come-PFV MEIYOU BU MA BA A
 (Intended:) ‘Did the newspaper arrive or not?’ (Zhu 1982: 203, adapted)
- c. Baozhi **lai-mei-lai** (ne) a?
 newspaper come-not.PFV-come NE A
 ‘Did the newspaper arrive or not?’

To clarify, the unacceptability of *a* with *ma* and *ba* does not derive from a phonological constraint that bans two consecutive identical vowels. We can see this in (92), where *a* is allowed when following a bipolar question that ends with an identical string bearing an identical tone as *ma*.

- (92) Xiaoming xiang-bu-xiang mama a?
 Xiaoming miss-not-miss mom A
 ‘Does Xiaoming miss his mom or not?’

The distribution of *a* can be naturally captured by positing that it semantically selects (nonempty) nonsingletons. If this division holds, it aligns perfectly with the observations in English (Biezma 2009) and in Thompson River Salish (Matthewson 2025) that bipolar questions are more compatible with the cornering move, a conse-

²² The adverbs *daodi* ‘on earth’ (Hsieh 2004, Law 2006, 2008, Her, Che & Bodomo 2022) and *jiujing* (M. Yuan 2015) have also been frequently used to illustrate a contrast between A-not-A and *ma* questions, with the claim that it can occur with the former but not the latter. However, our consultants uniformly report that they are perfectly acceptable with *ma*. This may reflect dialectal variation, but we also suggest that for speakers who allow *daodi* or *jiujing* with *ma*, it could be the case that some element in the sentence must be focused, thereby introducing a plural set of alternatives.

quence that follows from the generalization that the alternatives in the denotation of a bipolar question always exhaust the whole possibility space, as discussed in Section 5.3.

The second distributional fact concerns *nandao*, which triggers a negative bias toward the question nucleus (Xu 2017). It can occur with *ma* or RDs, but it cannot occur with *wh*-questions or other question(-like) markers (Huang, Li & Li 2009, Her, Che & Bodomo 2022), as shown in (93).

- (93) a. **Nandao* Zhangsan **weishenme** chi-le fan?
 NANDAO Zhangsan why eat-PFV meal
 b. *Nandao* Zhangsan chi-le fan **ma**/**meiyou*/**bu*/**ba*?
 NANDAO Zhangsan eat-PFV meal MA MEIYOU BU BA
 (Intended:) ‘Zhangsan didn’t eat, right?’ (Xu 2017: 13)
 c. **Nandao* Zhangsan **chi-mei-chi** fan (ne)?
 NANDAO Zhangsan eat-not.PFV-eat meal NE
 d. *Nandao* Zhangsan chi-le fan?
 NANDAO Zhangsan eat-PFV meal
 ‘Zhangsan didn’t eat, right?’ (Xu 2017: 13)

This distribution can be naturally captured by positing that *nandao* semantically selects singletons.²³ Such an account aligns with Xu’s (2017) view that *nandao* picks out a unique salient alternative.

Lastly, as again illustrated using a contrast between *ma* and A-not-A questions in (94–95), in Mandarin, only *ma* questions and *ba* declaratives can be felicitously answered with *dui*, *mei cuo*, or *shi de*, all of which roughly correspond to English *right* (Hsieh 2001, 2004, Guo 2002, Ye 2022, 2024).

- (94) A: Ni xihuan da qiu **ma**?
 2SG like play ball MA
 ‘Do you like playing basketball?’
 a. B: *Xihuan*. ‘I like it.’
 b. B: *Dui./Mei cuo./Shi de*. ‘Right.’ (Ye 2024: 8)
- (95) A: Ni **xi-bu-xihuan** da qiu?
 2SG like-not-like play ball
 ‘Do you like playing basketball?’
 a. B: *Xihuan*. ‘I like it.’
 b. B: #*Dui*./#*Mei cuo*./#*Shi de*. ‘Right.’ (Ye 2024: 6)

²³ Apparently, *nandao* can attach either to PerspP or to CommitP, and only in the former case can it embed. This view is supported by the observation that *ma* questions with *nandao* can be quasi-subordinated, whereas RDs with *nandao* cannot:

- (i) a. Xiaohong_i xiang zhidao [nandao ta_i de fenshu bi wo_j gao **ma**]_↑.
 Xiaohong want know NANDAO 3SG POSS score than 1SG high MA
 ‘Xiaohong_i wants to know the answer to this question: Her_i score is higher than mine_j, right?’
 b. *Xiaohong_i xiang zhidao [nandao ta_i de fenshu bi wo_j gao]_{↑↑}.
 Xiaohong want know NANDAO 3SG POSS score than 1SG high

While the semantics of response particles remain under debate (Farkas & Bruce 2010, Kramer & Rawlins 2011, Krifka 2013, Roelofsen & Farkas 2015, Holmberg 2016, Farkas & Roelofsen 2019), we follow Ye (2022) and treat Mandarin response particles as propositional anaphors, which presuppose a unique alternative that is the most salient. This way, a monopolar question can license ‘right’ when there is no other alternative of equal salience, and it is unlikely (if not impossible) for a bipolar question to be able to license ‘right’ because it has two equally salient alternatives.

To recapitulate, the distributions of *nandao*, *a*, and ‘right’ as a response, together with the empirical findings from the B-contexts, are summarized in Table 9. Interestingly, among polar question markers, there is a strong correlation: those that contain the negative element are bipolar, whereas the only one that does not (i.e., *ma*) is monopolar. On top of the distributional observations, this correlation further motivates the semantic implementations we propose in Table 6.

Table 9 Compatibility with *a*, B-contexts, *nandao*, and ‘right’ as a response

	<i>Wh-</i> Q_{PL}	<i>Meiyou</i> $\{p, \neg p\}$	A-not-A $\{p, \neg p\}$	<i>Bu</i> $\{p, \neg p\}$	<i>Ma</i> $\{p\}$	<i>Ne</i> Q_{PL}	<i>Ba</i> p
<i>A</i>	✓	✓	✓	✓	✗	✓	✗
B-contexts	✗	✗	✗	✗	✓	✗	✗
<i>Nandao</i>	✗	✗	✗	✗	✓	✗	✗
‘Right’ as a response	✗	✗	✗	✗	✓	✗	✓

6 Clause-typing above ForceP

In concluding our discussion of the full account for the embeddability and co-occurrence patterns of Mandarin question markers, we highlight a key point of departure between our framework and Dayal (2025) implementation of perspectival centering. Specifically, our approach allows perspective heads to type a propositional core into a question, rather than uniformly treating their at-issue content as an identity function that passes on a question already typed in a lower projection. We reproduce both implementations in (96) for comparison.

- (96) a. *Dayal’s (2025) Persp_{CQ}⁰*

$$\llbracket \text{Persp}_{CQ}^0 \rrbracket = [\lambda Q. \lambda x: \text{PR}(x). Q]$$
 (Dayal 2025: 673)
- b. *Denotation of ma*

$$\llbracket \text{ma} \rrbracket = [\lambda p. \lambda x: \text{PR}(x). [\lambda q. q = p]]$$

To fully articulate the motivation for this move, we revisit Dayal’s empirical foundation in Section 6.1, using her Hindi/Urdu data as an illustration. Section 6.2 demonstrates how Mandarin differs from Hindi/Urdu in terms of possible loci of clause-typing.

6.1 Hindi/Urdu polar question particle

In Hindi/Urdu, matrix interrogatives exhibit a characteristic questioning intonation, which we again use “↑” to represent, and they may optionally include the polar question particle (in the sense of Bhatt & Dayal 2014, 2020) *kya:*, as illustrated in (98) alongside the declarative base in (97).

- (97) anu ja:egi:.
Anu will.go
'Anu will go.'
- (98) [(**kya:**) anu ja:egi:ʔ]_↑
PQP Anu will.go
'Will Anu go?' (Dayal 2025: 665)

Embedded questions in Hindi/Urdu involve the complementizer *ki*, which is compatible with both declarative (99) and interrogative (100) complements and cannot be used in the matrix clause. When embedded, interrogative clauses bearing the questioning intonation, optionally with *kya:*, show the same sensitivity to the QS-NQS contrast as Mandarin *ma*.

- (99) anu ja:nti: hai [ki tum cai piyoge].
Anu know AUX SUB you tea will.drink
'Anu knows that you will drink tea.'
- (100) a. anu ja:nna: ca:hti: hai [ki (**kya:**) tum cai piyoge]_↑. QS: ✓
Anu to.know want AUX SUB PQP you tea will.drink
'Anu wants to know whether you will drink tea.'
- b. *anu ja:nti: hai [ki (**kya:**) tum cai piyoge]_↑. NQS: ✗
Anu know AUX SUB PQP you tea will.drink
Intended: 'Anu knows whether you will drink tea.'
- (Dayal 2025: 681)

Aside from quasi subordination, Hindi/Urdu allows regular question subordination that lacks questioning intonation, in which case a negative disjunct *ya: nahĩ:* 'or not' is obligatorily introduced (101). Note also that *ya: nahĩ:* may be used in matrix questions, but not without the questioning intonation (102).

- (101) ravi ja:nta: hai [ki anu ja:egi: *(**ya: nahĩ:**)].
Ravi know AUX SUB Anu will.go or not
(Intended:) 'Ravi knows whether Anu will go (or not).' (Dayal 2025: 697)
- (102) [anu ja:egi: **ya: nahĩ:**?]_{* (↑)}
Anu will.go or not
(Intended:) 'Will Anu go or not?'

A final and crucial note is that *kya:* and *ya: nahĩ:* can co-occur in both matrix and embedded contexts when the questioning intonation is present (103).

- (103) a. [(**kya:**) anu ja:egi: (**ya: nahĩ:**)?]_↑
PQP Anu will.go or not
'Will Anu go (or not)?'
- b. ravi ja:nna: ca:hta: hai [ki (**kya:**) anu ja:egi: (**ya: nahĩ:**)]_↑.
Ravi to.know want AUX SUB PQP Anu will.go or not
'Ravi wants to know whether Anu will go (or not).'
- (Dayal 2025: 696)

To account for (97–103), in addition to the three-way split in the left periphery and the p-activeness requirement introduced by PerspP, Dayal proposes that Hindi/Urdu CP (roughly corresponding to ForceP in our analysis) is underspecified for the [α Q] feature (or [α WH] in her terminology), and it is only activated as [+Q]

when embedded under $\text{Persp}_{\text{CQ}}^0$. Whereas a nonactivated $[\alpha\text{Q}]$ defaults to a propositional interpretation, an activated $[+\text{Q}]$ types the clause as a question and triggers a shift from a proposition to a set of propositions.

If we take (100) to establish that *kya*: surfaces at the PerspP level and (101) to establish that *ya*: *nahĩ*: surfaces at Dayal’s CP level, then given that *kya*: and *ya*: *nahĩ*: may co-occur while both of them are independently optional, Dayal’s account emerges as the most natural one, since the empirical landscape necessitates an analysis that allows for the possibility that the complement taken by the PerspP is already a question. When we turn to Mandarin, however, the data do not push for this necessity. Details follow in the next subsection.

6.2 Minimal structural domain

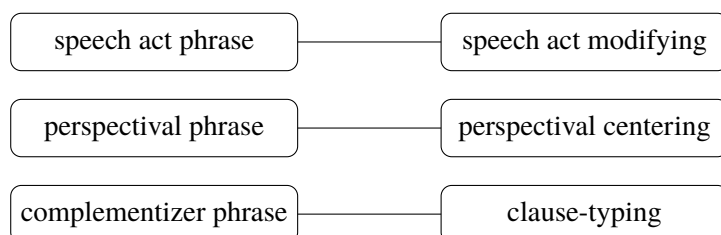
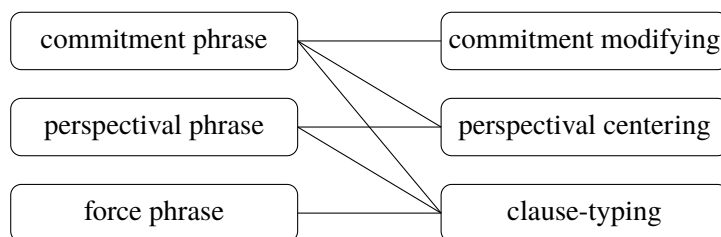
If one were to extend Dayal’s (2025) account for Hindi/Urdu to Mandarin, an empirical challenge arises concerning the mutual exclusivity of *bu* and *ma* with ForceP-level question markers, a pattern discussed in Section 2.1 and Section 4.1. We copy (12a) in (104a) and add data points for A-not-A in (104b) and for *wh*-questions in (104c), using *ma* as an example.

- (104) a. *Zuotian xia-le yu **meiyou** **ma**? meiyou + ma $\leadsto$ ✗
 yesterday fall-PFV rain MEIYOU MA
 Intended: ‘Did it rain yesterday?’
- b. *Zuotian **xia-mei-xia** yu **ma**? A-not-A + ma $\leadsto$ ✗
 yesterday fall-not.PFV-fall rain MA
 Intended: ‘Did it rain yesterday?’
- c. Zuotian **shenme shihou** xia-le yu **ma**? wh + ma $\leadsto$ ✓indefinite, ✗question
 yesterday what time fall-PFV rain MA
 ‘Did it rain sometime yesterday?’
 Not: ‘When did it rain yesterday?’

Examples (104a–104b) demonstrate that *ma* cannot select a clause already typed as a question, and (104c) demonstrates that *ma* cannot convert a clause that is underspecified for $[\alpha\text{Q}]$ into $[+\text{Q}]$. A more natural account, then, is to say that *bu* and *ma* type clauses.²⁴

As a consequence, taken together with evidence from Hindi/Urdu and Mandarin, our account points to a mapping in the left periphery between syntactic projections and semantic transformations that differs from Dayal’s one-to-one correspondence. It suggests that each semantic transformation has a *minimal* structural domain—ForceP for clause-typing, PerspP for perspectival centering, and CommitP for commitment modifying—and may take place at a projection higher than its minimal structural domain. We sketch this idea in (105).

²⁴ An alternative analysis where *bu* and *ma* are identity functions that select a phonologically null clause-typing head would require extra assumptions to explain their nonoptionality. One could also posit that *bu* and *ma* are force heads that obligatorily project to PerspP, or that they expone a fused force-perspective head. Without further empirical motivation, we set these possibilities aside.

(105) a. *Dayal's (2025) mapping*b. *Minimal structural domain*

7 Conclusion

The embeddability profiles of Mandarin question markers divide them into three groups: (i) the always embeddable ones; (ii) the restrictedly embeddable ones, which are sensitive to the PR; and (iii) the never embeddable ones. This trichotomy contradicts the traditional view on the syntactic statuses of these question markers that categorizes them using a root-nonroot distinction, highlighting a third possibility that warrants a conspiracy of syntax and semantics for the theory of complement selection. As such, *Dayal's (2025)* three-way split in the left periphery is corroborated to be systematically manifested in Mandarin: ForceP marks the earliest stage of clause-typing, PerspP triggers perspectival centering, and CommitP performs commitment modifying.

The non-co-occurrence of question markers is subject to the semantic types of their input and output rather than solely affected by a syntactic competition. When scrutinized under assertiveness diagnostics, they show a contrast that is characteristic of that between canonical assertions and questions, revealing that some seemingly homogeneous question markers are in fact different in their semantic nature. Aside from an assertion-question distinction, B-contexts and the distributional facts of various question-related elements motivate a monopolar-bipolar distinction. With these two distinctions established, a comprehensive account for the semantic interaction of Mandarin question markers is reached.

Hindi/Urdu and Mandarin show a contrast with respect to the possible loci of clause-typing, calling for a theory that puts a constraint only on the minimal size of the structural domain of a semantic transformation instead of forcing a one-to-one projection-function mapping. This insight provides crosslinguistic implications that shed light on our understanding of what happens where in the left periphery, as well as pointers for future inquiries about question structures and meanings.

References

AnderBois, Scott. 2011. *Issues and alternatives*. Santa Cruz: University of California, Santa Cruz dissertation.

- Aoun, Joseph & Yen-Hui Audrey Li. 1993a. On some differences between Chinese and Japanese *wh*-elements. *Linguistic Inquiry* 24(2). 365–372. <http://www.jstor.org/stable/4178815>.
- Aoun, Joseph & Yen-Hui Audrey Li. 1993b. *Wh*-elements in situ: Syntax or LF? *Linguistic Inquiry* 24(2). 199–238. <http://www.jstor.org/stable/4178811>.
- Asher, Nicholas & Brian Reese. 2005. Negative bias in polar questions. *Sinn und Bedeutung (SuB)* 9. 30–43. <https://doi.org/10.18148/SUB/2005.V9I0.716>.
- Asher, Nicholas & Brian Reese. 2007. Intonation and discourse: Biased questions. *Interdisciplinary Studies on Information Structure* 8. 1–38.
- Bach, Kent & Robert Harnish. 1979. *Linguistic communication and speech acts*. Cambridge, MA: MIT Press.
- Bailey, Laura. 2010. Sentential word order and the syntax of question particles. *Newcastle Working Papers in Linguistics* 16. 23–43.
- Bailey, Laura. 2013. *The syntax of question particles*. Newcastle upon Tyne: Newcastle University dissertation.
- Belletti, Adriana. 2004. Aspects of the low IP area. In Luigi Rizzi (ed.), *The structure of CP and IP: The cartography of syntactic structures, volume 2*, 16–51. New York: Oxford University Press. <https://doi.org/10.1093/oso/9780195159486.003.0002>.
- Beltrama, Andrea, Erlinde Meertens & Maribel Romero. 2020. Alternative questions: Distinguishing between negated and complementary disjuncts. *Semantics and Pragmatics* 13, 5. <https://doi.org/10.3765/sp.13.5>.
- Bhatt, Rajesh & Veneeta Dayal. 2014. Polar *kya*: Y/N or speech act operator. Paper presented at the Workshop on Non-Canonical Questions and Interface Issues, University of Konstanz, 17–19 February.
- Bhatt, Rajesh & Veneeta Dayal. 2020. Polar question particles: Hindi-Urdu *kya*:. *Natural Language & Linguistic Theory* 38(4). 1115–1144. <https://doi.org/10.1007/s11049-020-09464-0>.
- Biberauer, Theresa, Anders Holmberg & Ian Roberts. 2014. A syntactic universal and its consequences. *Linguistic Inquiry* 45(2). 169–225. https://doi.org/10.1162/LING_a_00153.
- Biezma, Maria. 2009. Alternative vs polar questions: The *cornering* effect. *Semantics and Linguistic Theory (SALT)* 19. 37–54. <https://doi.org/10.3765/salt.v19i0.2519>.
- Biezma, María & Kyle Rawlins. 2012. Responding to alternative and polar questions. *Linguistics and Philosophy* 35(5). 361–406. <https://doi.org/10.1007/s10988-012-9123-z>.
- Bolinger, Dwight. 1979. Yes–no questions are not alternative questions. In Henry Hiz (ed.), *Questions*, 87–105. Dordrecht: Reidel. https://doi.org/10.1007/978-94-009-9509-3_3.
- Brasoveanu, Adrian. 2013. Modified numerals as post-suppositions. *Journal of Semantics* 30(2). 155–209. <https://doi.org/10.1093/jos/ffs003>.
- Burling, Robbins. 1973. *English in black and white*. New York: Holt, Rinehart & Winston.
- Chao, Yuen-Ren. 1926. Beijing, Suzhou, Changzhou yu zhuci de yanjiu [A study on Particles in Beijing, Suzhou and Changzhou]. *Qinghua Daxue Xuebao (Kexue Jishu Ban) [Journal of Tsinghua University (Science and Technology)]* 3(2). 865–917. <https://doi.org/10.16511/j.cnki.qhdxxb.1926.02.001>.
- Chao, Yuen-Ren. 1965. *A grammar of spoken Chinese*. Berkeley, CA: University of California Press.
- Chappell, Hilary. 1991. Strategies for the assertion of obviousness and disagreement in Mandarin: A semantic study of the modal particle *me*. *Australian Journal of Linguistics* 11(1). 39–65. <https://doi.org/10.1080/07268609108599451>.
- Charlow, Simon. To appear. Post-suppositions and semantic theory. *Journal of Semantics*.
- Cheng, Lisa Lai-Shen. 1991. *On the typology of wh-questions*. Cambridge, MA: Massachusetts Institute of Technology dissertation.

- Cheung, Lawrence Yam-Leung. 2009. Dislocation focus construction in Chinese. *Journal of East Asian Linguistics* 18(3). 197–232. <https://doi.org/10.1007/s10831-009-9046-z>.
- Chu, Chauncey Cheng-Hsi. 1998. *A discourse grammar of Mandarin Chinese*. New York: Peter Lang.
- Chu, Chauncey Cheng-Hsi. 2009. Relevance and the discourse functions of Mandarin utterance-final modality particles. *Language and Linguistics Compass* 3(1). 282–299. <https://doi.org/10.1111/j.1749-818X.2008.00105.x>.
- Chu, Chauncey Cheng-Hsi & Bin Li. 2004. Lun Xiandai Hanyu ju mo qingtai xuci ji qi Ying yi—yi “ba” de yu pian gongneng wei li [Translating Modern Chinese utterance-final particles into English: A case study of the discourse function of BA]. *Waiyu Xuekan [Foreign Language Research]* 27(6). 1–10. <https://doi.org/10.16263/j.cnki.23-1071/h.2004.06.001>.
- Coniglio, Marco & Iulia Zegrean. 2012. Splitting up force: Evidence from discourse particles. In Lobke Aelbrecht, Liliane Haegeman & Rachel Nye (eds.), *Main clause phenomena: New horizons*, 229–256. Amsterdam: John Benjamins. <https://doi.org/10.1075/la.190.10con>.
- Constant, Noah. 2014. *Contrastive topic: Meanings and realizations*. Amherst, MA: University of Massachusetts dissertation. <https://doi.org/10.7275/5694973.0>.
- Davies, William & Stanley Dubinsky. 1998. Sentential subjects as complex NPs: New reasons for an old account of subjacency. *Chicago Linguistic Society (CLS)* 34(1). 83–94.
- Dayal, Veneeta. 2025. The interrogative left periphery: How a clause becomes a question. *Linguistic Inquiry* 56(4). 663–712. https://doi.org/10.1162/ling_a_00507.
- Dayal, Veneeta & Jane Grimshaw. 2009. Subordination at the interface: The quasi-subordination hypothesis. Ms., Rutgers University, New Brunswick, NJ.
- Ding, Shengshu, Shuxiang Lü, Rong Li, Dexuan Sun, Xiechu Guan, Jing Fu, Shengzhang Huang & Zhiwen Chen. 1961. *Xiandai Hanyu yufa jianghua [Lectures on Modern Chinese grammar]*. Beijing: Commercial Press.
- Emonds, Joseph. 1970. *Root and structure-preserving transformations*. Cambridge, MA: Massachusetts Institute of Technology dissertation.
- Emonds, Joseph. 1988. *A transformational approach to English syntax: Root, structure-preserving and local transformations*. New York: Academic.
- Emonds, Joseph. 2004. Unspecified categories as the key to root constructions. In David Adger, Cécile de Cat & George Tsoulas (eds.), *Peripheries: Syntactic edges and their effects*, 75–120. Dordrecht: Springer. https://doi.org/10.1007/1-4020-1910-6_4.
- Erlewine, Michael Yoshitaka. 2017. Low sentence-final particles in Mandarin Chinese and the Final-over-Final Constraint. *Journal of East Asian Linguistics* 26(1). 37–75. <https://doi.org/10.1007/s10831-016-9150-9>.
- Ernst, Thomas. 1994. Conditions on Chinese A-not-A questions. *Journal of East Asian Linguistics* 3(3). 241–264. <https://doi.org/10.1007/BF01733065>.
- Ettinger, Allyson & Sophia Malamud. 2019. Mandarin utterance-final particle *ba* () in the conversational scoreboard. *Sinn und Bedeutung (SuB)* 19. 232–251. <https://doi.org/10.18148/SUB/2015.V19I0.840>.
- Farkas, Donka & Kim Bruce. 2010. On reacting to assertions and polar questions. *Journal of Semantics* 27(1). 81–118. <https://doi.org/10.1093/jos/ffp010>.
- Farkas, Donka & Floris Roelofsen. 2019. Polarity particles revisited. *Semantics and Pragmatics* 12. <https://doi.org/10.3765/sp.12.15>.
- Guo, Rui. 2002. Ma wenju de quexindu he huida fangshi [The degree of credibility of and the answers to “ma”-questions]. *Shijie Hanyu Jiaoxue [Chinese Teaching in the World]* 2. 13–23.

- Hagstrom, Paul. 2006. A-not-A questions. In Martin Everaert & Henk van Riemsdijk (eds.), *The Blackwell companion to syntax*, vol. 1, 173–213. Oxford: Blackwell. <https://doi.org/10.1002/9780470996591.ch7>.
- Hamblin, Charles Leonard. 1973. Questions in Montague English. *Foundations of Language* 10(1). 41–53. <http://www.jstor.org/stable/25000703>.
- Han, Yang Saxena. 1995. A pragmatic analysis of the *ba* particle in Mandarin Chinese. *Journal of Chinese Linguistics* 23(2). 99–128. <http://www.jstor.org/stable/23756541>.
- Hartman, Jeremy. 2012. *Varieties of clausal complementation*. Cambridge, MA: Massachusetts Institute of Technology dissertation.
- Henry, Alison. 1995. *Belfast English and standard English: Dialect variation and parameter setting*. New York: Oxford University Press.
- Her, One-Soon, Dewei Che & Adams Bodomo. 2022. On a dichotomy of question types: the case of Mandarin Chinese and Changsha Xiang. *The Linguistic Review* 39(2). 257–291. <https://doi.org/10.1515/tlr-2022-2087>.
- Holmberg, Anders. 2016. *The Syntax of Yes and No*. Oxford: Oxford University Press, Incorporated.
- Hooper, Joan & Sandra Thompson. 1973. On the applicability of root transformations. *Linguistic Inquiry* 4(4). 465–497. <http://www.jstor.org/stable/4177789>.
- Hsieh, Miao-Ling. 2001. *Form and meaning: Negation and question in Chinese*. Los Angeles: University of Southern California dissertation.
- Hsieh, Miao-Ling. 2004. On the Licensing of A-not-A Forms in Chinese. *Concentric: Studies in Linguistics* 30. 68–92.
- Hu, Mingyang. 1981a. Beijing hua de yuqi zhuci he tanci (shang) [Mood particles and interjections in Beijing dialect (part 1)]. *Zhongguo Yuwen [Studies of the Chinese Language]* 19(5). 347–350.
- Hu, Mingyang. 1981b. Beijing hua de yuqi zhuci he tanci (xia) [Mood particles and interjections in Beijing dialect (part 2)]. *Zhongguo Yuwen [Studies of the Chinese Language]* 19(6). 416–423.
- Huang, Cheng-Teh James. 1982a. *Logical relations in Chinese and the theory of grammar*. Cambridge, MA: Massachusetts Institute of Technology dissertation.
- Huang, Cheng-Teh James. 1982b. Move WH in a language without WH movement. *The Linguistic Review* 1(4). 369–416. <https://doi.org/10.1515/tlir.1982.1.4.369>.
- Huang, Cheng-Teh James. 1991. Modularity and Chinese A-not-A questions. In Carol Georgopoulos & Roberta Ishihara (eds.), *Interdisciplinary approaches to language: Essays in honor of S.-Y. Kuroda*, 305–332. Dordrecht: Springer. https://doi.org/10.1007/978-94-011-3818-5_16.
- Huang, Cheng-Teh James, Yen-Hui Audrey Li & Yafei Li. 2009. *The syntax of Chinese*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9781139166935>.
- Kamali, Beste. 2020. Motivating polar and focal alternatives in polar questions in broad focus. Ms., Leibniz-Zentrum Allgemeine Sprachwissenschaft, Berlin. <https://ling.auf.net/lingbuzz/005477>.
- Kamali, Beste. 2025. Evidential bias across clause types. In Tue Trinh, Anton Benz, Daniel Goodhue, Kazuko Yatsushiro & Manfred Krifka (eds.), *Biased questions: Experimental results and theoretical modelling*. Berlin: Language Science Press. <https://doi.org/10.5281/ZENODO.17084327>.
- Karttunen, Lauri. 1977. Syntax and semantics of questions. *Linguistics and Philosophy* 1(1). 3–44. <https://doi.org/10.1007/BF00351935>.
- Kim, Juhyae. 2019. *Speech act intensification in Mandarin*. Swarthmore, PA: Swarthmore College BA thesis.
- Kramer, Ruth & Kyle Rawlins. 2011. Polarity particles: An ellipsis account. In Suzi Lima, Kevin Mullin, & Brian Smith (eds.), *North East Linguistic Society (NELS)*. 479–492.

- Krifka, Manfred. 2013. Response particles as propositional anaphors. *Semantics and Linguistic Theory (SALT)* 23. 1–18. <https://doi.org/10.3765/salt.v23i0.2676>.
- Krifka, Manfred. 2015. Bias in commitment space semantics: Declarative questions, negated questions, and question tags. *Semantics and Linguistic Theory (SALT)* 25. 328–345. <https://doi.org/10.3765/salt.v25i0.3078>.
- Krifka, Manfred. 2022. Modelling questions in Commitment Spaces. In Moritz Cordes (ed.), *Asking and answering: Rivaling approaches to interrogative methods*, 63–95. Tübingen: Narr Francke Attempto. <https://doi.org/10.24053/9783823394808>.
- Krifka, Manfred. 2024. Performative updates and the modeling of speech acts. *Synthese* 203(1). <https://doi.org/10.1007/s11229-023-04359-0>.
- Law, Paul. 2006. Adverbs in A-not-A questions in Mandarin Chinese. *Journal of East Asian Linguistics* 15(2). 97–136. <https://doi.org/10.1007/s10831-005-4916-5>.
- Law, Paul. 2008. The *wh/q*-polarity adverb *daodi* in Mandarin Chinese and the syntax of focus. *The Linguistic Review* 25(3–4). 297–345. <https://doi.org/10.1515/TLIR.2008.009>.
- Lee, Hun-Tak Thomas. 1986. *Studies on quantification in Chinese*. Los Angeles: University of California dissertation.
- Lee, Tommy Tsz-Ming. 2017. Defocalization in Cantonese right dislocation. *Gengo Kenkyu* 152. 59–87. https://doi.org/10.11435/gengo.152.0_59.
- Lees, Robert. 1960. *Grammar of English nominalizations: International Journal of American Linguistics* V26, No. 3, Part 2, July, 1960. Bloomington, IN: Indiana University, Research Center for Anthropology, Folklore, & Linguistics.
- Li, Boya. 2006. *Chinese final particles and the syntax of the periphery*. Leiden: Leiden University dissertation.
- Li, Charles Na & Sandra Thompson. 1979. The pragmatics of two types of yes-no questions in Mandarin and its universal implications. *Chicago Linguistic Society (CLS)* 15(1). 197–206.
- Li, Charles Na & Sandra Thompson. 1981. *Mandarin Chinese: A functional reference grammar*. Berkeley, CA: University of California Press.
- Li, Yen-Hui Audrey. 1992. Indefinite *Wh* in Mandarin Chinese. *Journal of East Asian Linguistics* 1(2). 125–155. <https://doi.org/10.1007/BF00130234>.
- Liu, Fang & Yi Xu. 2005. Parallel encoding of focus and interrogative meaning in Mandarin intonation. *Phonetica* 62(2–4). 70–87. <https://doi.org/10.1159/000090090>.
- Liu, Mau-Tsai. 1964. *Deutsch-Chinesische Syntax: Ein praktisches Handbuch der modernen chinesischen Umgangssprache [German-Chinese Syntax: A practical handbook of Modern Chinese colloquial language]*. Berlin: Walter de Gruyter. <https://doi.org/10.1515/9783110834154>.
- Liu, Yuyang. 2025. Mandarin Chinese *ma*: Q morpheme, SA intensifier, or PQP? *North East Linguistic Society (NELS)* 55(2). 89–98.
- Lü, Shuxiang. 1980. *Xiandai Hanyu babai ci [Eight hundred words in Modern Chinese]*. Beijing: Commercial Press.
- Lü, Shuxiang. 1982. *Zhongguo wenfa yaolüe [Essentials of Chinese Grammar]*. Beijing: Commercial Press.
- Luo, Yitong. To appear. Embedded *ma*: New data and analysis for Mandarin polar questions. In *Proceedings of the 36th North American Conference on Chinese Linguistics (NACCL)*.
- Matthewson, Lisa. 2025. Polar questions in *neʔkepmxc[n]*: Monopolar, bipolar, and exhaustive. *Natural Language Semantics*. <https://doi.org/10.1007/s11050-025-09239-6>.

- McCloskey, James. 2006. Questions and questioning in a local English. In Raffaella Zanuttini, Héctor Campos, Elena Herburger & Paul Portner (eds.), *Crosslinguistic research in syntax and semantics: Negation, tense, and clausal architecture*, 87–126. Washington, DC: Georgetown University Press.
- Miyagawa, Shigeru. 2022. *Syntax in the treetops*. Cambridge, MA: MIT Press. <https://doi.org/10.7551/mitpress/14421.001.0001>.
- Miyagawa, Shigeru & Virginia Hill. 2025. Commitment Phrase: Linking proposition to illocutionary force. *Linguistic Inquiry* 56(3). 589–604. https://doi.org/10.1162/ling_a_00503.
- Pan, Victor Junnan. 2015. Mandarin peripheral construals at the syntax-discourse interface. *The Linguistic Review* 32(4). 819–868. <https://doi.org/10.1515/tlr-2015-0005>.
- Pan, Victor Junnan. 2017. Optional projections in the left-periphery in Mandarin Chinese. In Fuzhen Si (ed.), *Studies on syntactic cartography*, 216–248. Beijing: China Social Sciences Press.
- Pan, Victor Junnan. 2018. Derivation of the apparent narrow scope of sentence-final particles in Chinese: A reply to Erlewine (2017). *Studies in Chinese Linguistics* 39(2). 99–125. <https://doi.org/10.2478/scl-2018-0004>.
- Pan, Victor Junnan. 2019. *Architecture of the periphery in Chinese: Cartography and minimalism*. London: Routledge. <https://doi.org/10.4324/9781315158228>.
- Pan, Victor Junnan. 2022. Deriving head-final order in the peripheral domain of Chinese. *Linguistic Inquiry* 53(1). 121–154. https://doi.org/10.1162/ling_a_00396.
- Pan, Victor Junnan & Waltraud Paul. 2016. Why Chinese SFPs are neither optional nor disjunctors. *Lingua* 170. 23–34. <https://doi.org/10.1016/j.lingua.2015.10.005>.
- Paul, Waltraud. 2005. Low IP area and left periphery in Mandarin Chinese. *Recherches Linguistiques de Vincennes* 33. 111–134. <https://doi.org/10.4000/rlv.1303>.
- Paul, Waltraud. 2014. Why particles are not particular: Sentence-final particles in Chinese as heads of a split CP. *Studia Linguistica* 68(1). 77–115. <https://doi.org/10.1111/stul.12020>.
- Paul, Waltraud. 2015. *New perspectives on Chinese syntax*. Berlin: De Gruyter Mouton. <https://doi.org/10.1515/9783110338775>.
- Paul, Waltraud & Victor Junnan Pan. 2016. What you see is what you get: Chinese sentence-final particles as head-final complementizers. In Josef Bayer & Volker Struckmeier (eds.), *Discourse particles: Formal approaches to their syntax and semantics*, 49–77. Berlin: De Gruyter. <https://doi.org/10.1515/9783110497151-003>.
- Reese, Brian & Nicholas Asher. 2007. Prosody and the interpretation of tag questions. *Sinn und Bedeutung (SuB)* 11. 448–462. <https://doi.org/10.18148/SUB/2007.V11I0.657>.
- Reese, Brian & Nicholas Asher. 2009. Biased questions, intonation, and discourse. In Malte Zimmermann & Caroline Féry (eds.), *Information structure: Theoretical, typological, and experimental perspectives*, 139–174. New York: Oxford University Press. <https://doi.org/10.1093/acprof:oso/9780199570959.003.0007>.
- Rizzi, Luigi. 1997. The fine structure of the left periphery. In Liliane Haegeman (ed.), *Elements of grammar: Handbook in generative syntax*, 281–337. Dordrecht: Springer. https://doi.org/10.1007/978-94-011-5420-8_7.
- Rizzi, Luigi. 2001. On the position “Int(errogative)” in the left periphery of the clause. In Guglielmo Cinque & Giampaolo Salvi (eds.), *Current studies in Italian syntax: Essays offered to Lorenzo Renzi*, 287–296. Bradford: Emerald. https://doi.org/10.1163/9780585473949_016.

- Rizzi, Luigi. 2004. Locality and left periphery. In Adriana Belletti (ed.), *Structures and beyond: The cartography of syntactic structures, volume 3*, 223–251. New York: Oxford University Press. <https://doi.org/10.1093/oso/9780195171976.003.0008>.
- Roberts, Craige. 2012. Information structure in discourse: Towards an integrated formal theory of pragmatics. *Semantics and Pragmatics* 5. <https://doi.org/10.3765/sp.5.6>.
- Roberts, Craige. 2023. Imperatives in dynamic pragmatics. *Semantics and Pragmatics* 16(7). <https://doi.org/10.3765/sp.16.7>.
- Roelofsen, Floris & Donka Farkas. 2015. Polarity particle responses as a window onto the interpretation of questions and assertions. *Language* 91(2). 359–414. <https://doi.org/10.1353/lan.2015.0017>.
- van Rooij, Robert & Marie Šafářová. 2003. On polar questions. *Semantics and Linguistic Theory (SALT)* 13. 292–309. <https://doi.org/10.3765/salt.v13i0.2887>.
- Rosenbaum, Peter. 1965. *The grammar of English predicate complement constructions*. Cambridge, MA: Massachusetts Institute of Technology dissertation.
- Rudin, Deniz. 2022. Intonational commitments. *Journal of Semantics* 39(2). 339–383. <https://doi.org/10.1093/jos/ffac002>.
- Sadock, Jerrold. 1971. Queclaratives. *Chicago Linguistic Society (CLS)* 7. 223–231.
- Sadock, Jerrold. 1974. *Toward a linguistic theory of speech acts*. New York: Academic.
- Schaffar, Wolfram & Lansun Chen. 2001. Yes–no questions in Mandarin and the theory of focus. *Linguistics* 39(5). 837–870. <https://doi.org/10.1515/ling.2001.036>.
- Searle, John. 1979. *Expression and meaning: Studies in the theory of speech acts*. Cambridge: Cambridge University Press. <https://doi.org/10.1017/CBO9780511609213>.
- Soh, Hooi Ling. 2005. *Wh-in-situ in Mandarin Chinese*. *Linguistic Inquiry* 36(1). 143–155. <https://doi.org/10.1162/ling.2005.36.1.143>.
- Tabatowski, Michael. 2022. *Preferring to learn: An attitudinal approach to polar questions*. Chicago: The University of Chicago dissertation.
- Takahashi, Shōichi. 2010. The hidden side of clausal complements. *Natural Language & Linguistic Theory* 28(2). 343–380. <https://doi.org/10.1007/s11049-010-9091-3>.
- Tang, Ting-Chi. 1988. *Hanyu cifa jufa lunji [Studies on Chinese morphology and syntax]*. Taipei: Taiwan Student Book.
- Tsai, Wei-Tien Dylan. 1994. *On economizing the theory of A-bar dependencies*. Cambridge, MA: Massachusetts Institute of Technology dissertation.
- Tsai, Wei-Tien Dylan & Ching-Yu Helen Yang. To appear. On the inner-outer dichotomy of A-not-A questions: A minimalist-cartographic account. In Andrew Simpson (ed.), *New insights into theoretical syntax from Asian languages: Studies in honor of C.-T. James Huang*, 264–282. Amsterdam: John Benjamins.
- Williamson, Timothy. 1996. Knowing and asserting. *The Philosophical Review* 105(4). 489–523. <https://doi.org/10.2307/2998423>.
- Woods, Rebecca. 2016. *Investigating the syntax of speech acts: Embedding illocutionary force*. York: University of York dissertation.
- Wu, Guo. 2005. The discourse function of the Chinese particle *NE* in statements. *Journal of the Chinese Language Teachers Association* 40(1). 47–82.
- Xu, Beibei. 2017. *Question bias and biased question words in Mandarin, German and Bangla*. New Brunswick, NJ: Rutgers University dissertation.

- Yang, Yu'an. 2021. Represent an issue as open: Mandarin discourse particle *ba*. *Semantics and Linguistic Theory (SALT)* 30. 523–541. <https://doi.org/10.3765/salt.v30i0.4832>.
- Ye, Shumian. 2022. Xiandai Hanyu quezheng yingda ci de yuyi [The meaning of the confirmative response words in Chinese]. *Yufa Yanjiu he Tansuo [Grammatical research and exploration]* 21. 177–195.
- Ye, Shumian. 2024. Pianxiang ji xing wen ju de dongtai yuyi yanjiu [Dynamic semantics of biased polar questions in Mandarin]. Ms., Peking University.
- Yip, Ka-Fai. 2025. A unified biclausal approach to right dislocation in Chinese. *Journal of East Asian Linguistics*. <https://doi.org/10.1007/s10831-025-09307-2>.
- Yuan, Mengxi. 2015. *Chinese discourse adverbs as presupposition triggers*. Hong Kong: City University of Hong Kong dissertation.
- Yuan, Mengxi & Yurie Hara. 2019. Xiandai Hanyu “ma” wen ju yu “A bu A” wen ju de yuyi chayi [The semantic distinction between Chinese *ma* questions and A-not-A questions]. *Dangdai Yuyanxue [Contemporary Linguistics]* 21(1). 43–67.
- Yuan, Xuetong. 2020. Challenging the presuppositions of questions: The case of *ba*-interrogatives. *Sinn und Bedeutung (SuB)* 24(2). 469–484. <https://doi.org/10.18148/SUB/2020.V24I2.910>.
- Yuan, Xuetong. 2021. Extracting commitments: Rising intonation and *ba* in Mandarin. *Sinn und Bedeutung (SuB)* 25. 908–923. <https://doi.org/10.18148/SUB/2021.V25I0.974>.
- Yuan, Xuetong. 2023. Updating unexpected moves. *Sinn und Bedeutung (SuB)* 27. 748–760. <https://doi.org/10.18148/SUB/2023.V27.1098>.
- Zhu, Dexi. 1982. *Yufa jiangyi [Lectures on grammar]*. Beijing: Comercial Press.
- Zhu, Dexi. 1985. Hanyu fangyan de liang zhong fanfu wen ju [Two types of positive-negative questions in Chinese dialects]. *Zhongguo Yuwen [Studies of the Chinese Language]* 23(1). 10–20.

Yuyang Liu
Department of Linguistics
Yale University
37 Hillhouse Ave
New Haven, CT 06511
yuyang.liu@yale.edu

Yitong Luo
Department of Linguistics
Yale University
37 Hillhouse Ave
New Haven, CT 06511
yitong.luo@yale.edu